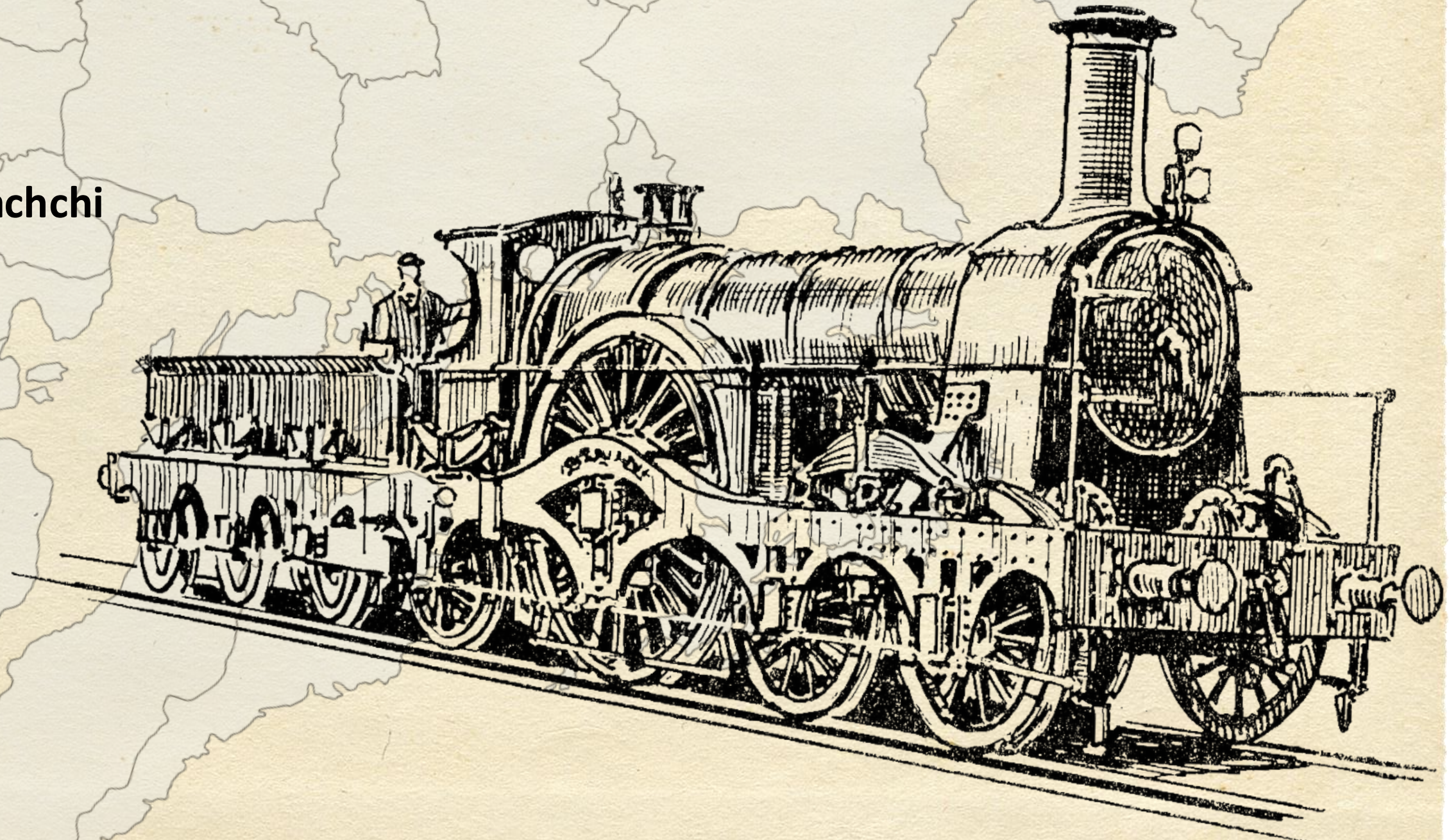
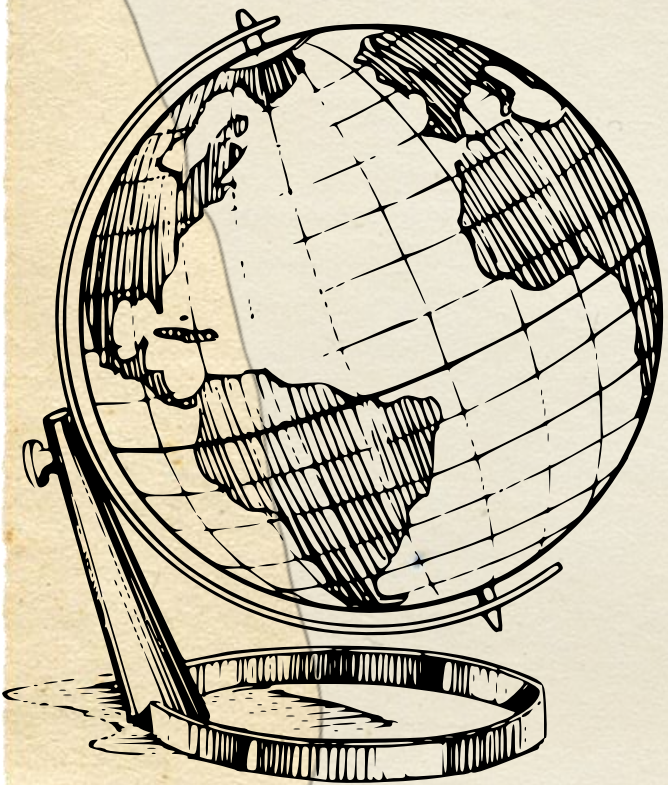


CeylonRail Intelligence

AI-Driven Intelligent Railway Operations System for Sri Lanka

Project ID: R26-IT-127

Supervisor: Ms. Chathurangika Kahandawaarachchi



AI-Driven Intelligent Railway Operations System for Sri Lanka

Our Team



IT22217240

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AI-Driven Intelligent Railway Operations System for Sri Lanka

Proposal Feedback and Improvements

Proposal Comment	How We Addressed It
Limited understanding of the research problem and methodology	Improved understanding of the overall railway AI system, research objectives, and component methodologie
Literature review was not clearly presented	Conducted and summarized literature review related to railway forecasting, AI analytics, and intelligent transportation systems
Clear system architecture/diagram was missing	Designed and added a complete system architecture diagram for the overall solution
Individual components were not clearly explained	Clearly separated the project into forecasting, delay prediction, fraud detection, and scheduling optimization components
Need better presentation and communication of the solution	Improved presentation structure with organized slides, visuals, dashboards, graphs, and prototype evidence

AI-Driven Intelligent Railway Operations System for Sri Lanka

Project Overview

Research Problem

- Overcrowding during peak travel periods
- Inaccurate passenger demand estimation
- Delays and schedule inefficiencies
- Manual fraud detection processes
- Poor railway capacity planning

Research Gap

- Existing systems lack AI-driven railway analytics
- Limited real-time forecasting capabilities
- No adaptive seat allocation based on predicted demand
- Lack of integrated intelligent railway decision-support systems

Existing Solutions :

- Traditional railway scheduling systems
- Manual passenger demand estimation
- Static seat allocation methods
- Basic railway monitoring systems



AI-Driven Intelligent Railway Operations System for Sri Lanka

Project Overview

Proposed Solution :

- AI-based passenger demand forecasting
- Adaptive seat allocation
- Probabilistic delay prediction
- AI-based fraud detection
- Multi-objective scheduling optimization

Design Excellence :

- AI-driven railway analytics and intelligent decision support
- Integrated AI-based railway optimization
- Real-time passenger demand forecasting and adaptive seat allocation
- Centralized intelligent dashboard for railway monitoring and operational planning

Who Will Be Benefited by the Solution		
Beneficiary		Benefit
	Sri Lanka Railways	 Improved operational planning and resource management
	Railway Passengers	 Reduced overcrowding and better travel experience
	Railway Administrators	 Intelligent decision-support and railway analytics
	Transportation Authorities	 Data-driven railway monitoring and optimization
	Daily Commuters	 Improved seat allocation and service reliability



Overall System Design

Excellence

CapCut

16:09

being one of the most important and

Multi-Objective Train Scheduling Optimization

IT22217240 - Narasinghe N.K.B.N.N

R26-IT-127



Multi-Objective Train Scheduling Optimization

How the Solution Addresses the Problem:

- Lack of optimized train scheduling
- Overcrowding during peak hours
- Poor resource utilization
- Difficulty balancing multiple objectives

User Requirements Addressed:

- NSGA-II optimization framework
- Multi-objective train scheduling
- Capacity management optimization
- Delay & congestion reduction

Key Features:

- Pareto-optimal schedule generation
- Peak-hour demand balancing
- Train & coach allocation optimization
- Simulation-based scenario testing



Multi-Objective Train Scheduling Optimization

Design Excellence / Contribution:

Multi-Objective Optimization Intelligence

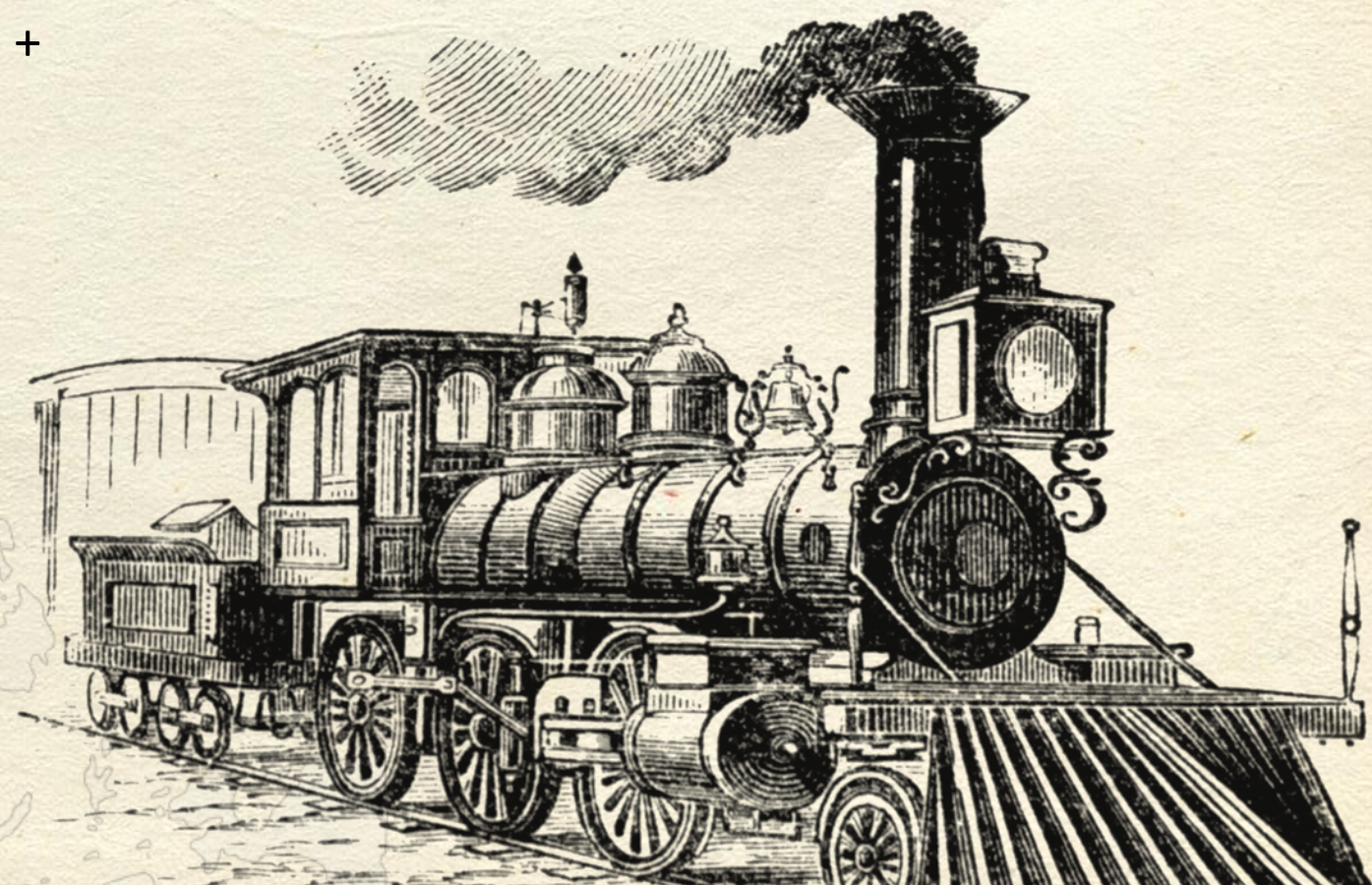
- Integrated scheduling & capacity optimization using NSGA-II
- Generates balanced trade-off railway schedules
- Improves efficiency, robustness, and passenger comfort

User Feedback:

- Better crowd management is highly valuable
- Reduced delays improve passenger experience
- Visualization dashboards improve railway decision-making
- Multi-scenario simulation helps operational planning

Prototype Implementation:

- Frontend Prototype (React + Tailwind + Leaflet)
- Backend Prototype (Node.js + Express + MongoDB)
- Optimization Engine (Python + NSGA-II + Simulation)



Prototype

```
[19] ipip install simpy
Requirement already satisfied: simpy in /usr/local/lib/python3.12/dist-packages (4.1.1)

[20] import simpy
import random
import pandas as pd
import numpy as np

[21] df = pd.read_excel(
    '/content/sri_lanka_railway_scheduling_dataset.xlsx'
)
df.head()
```

	Train_ID	Origin_Station	Destination_Station	Train_Type	Departure_Time	Arrival_Time	Passenger_Count	Capacity
0	TR1328	Colombo Fort	Galle	Intercity	2025-01-13 12:15:00	2025-01-13 17:15:00	332	700
1	TR1288	Maradana	Badulla	Night Mail	2025-11-29 21:45:00	2025-11-30 02:45:00	382	600
2	TR1053	Colombo Fort	Galle	Normal	2025-07-14 07:30:00	2025-07-14 14:30:00	582	800
3	TR1296	Maradana	Badulla	Normal	2025-12-27 06:00:00	2025-12-27 11:00:00	249	200

```
[17] pareto_df.to_csv(
    'pareto_solutions.csv',
    index=False
)

print("Saved Successfully!")

Saved Successfully!

[18] import plotly.express as px

fig = px.scatter(
    pareto_df,
    x='Delay',
    y='Operational_Cost',
    color='Overcrowding',
    title='Pareto Front Analysis'
)

fig.show()
```

Pareto Front Analysis



```
[25] env.run(until=30)

Time: 0
Train: TR101
Route: Colombo -> Kandy

Passengers: 571
Capacity: 500

Delay: 29 mins
Weather: Sunny

Overcrowded: True

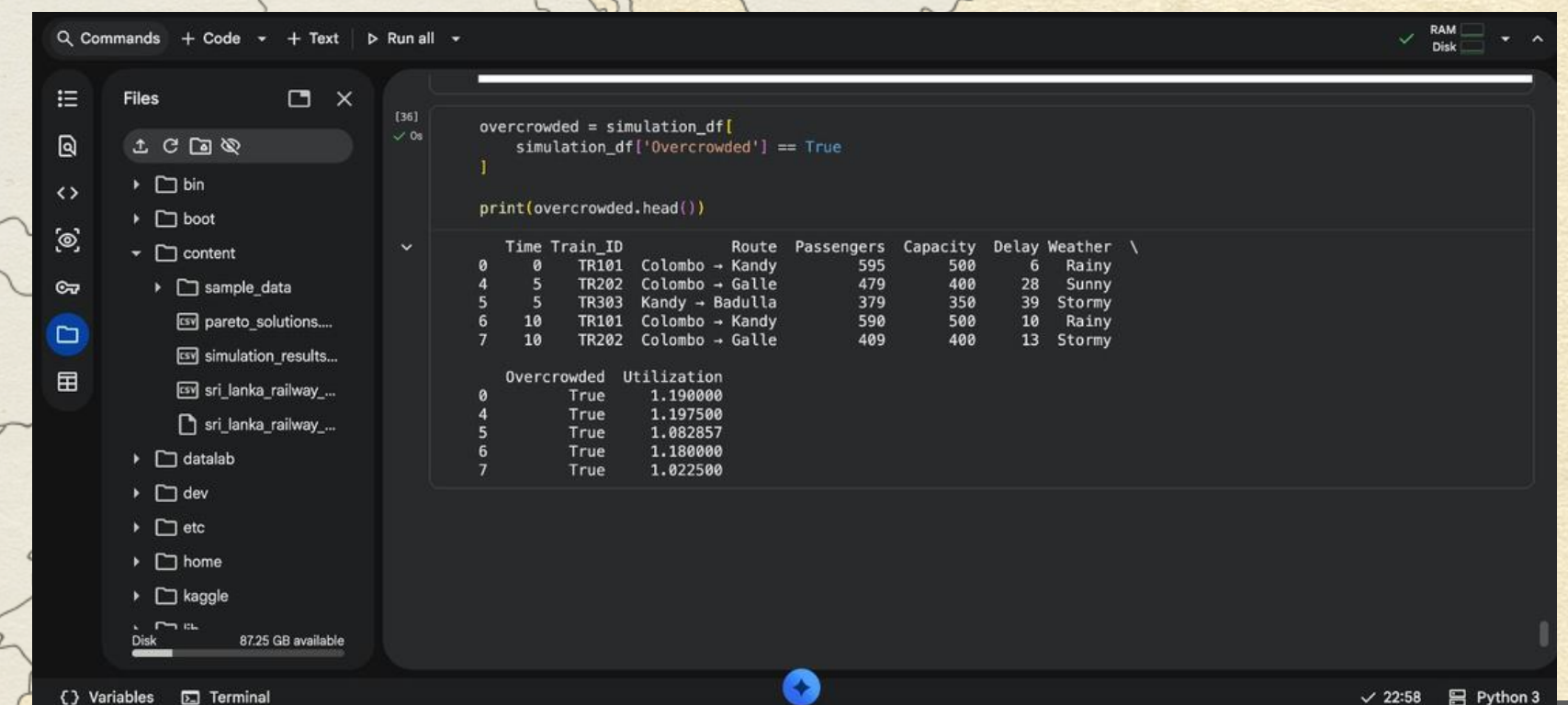
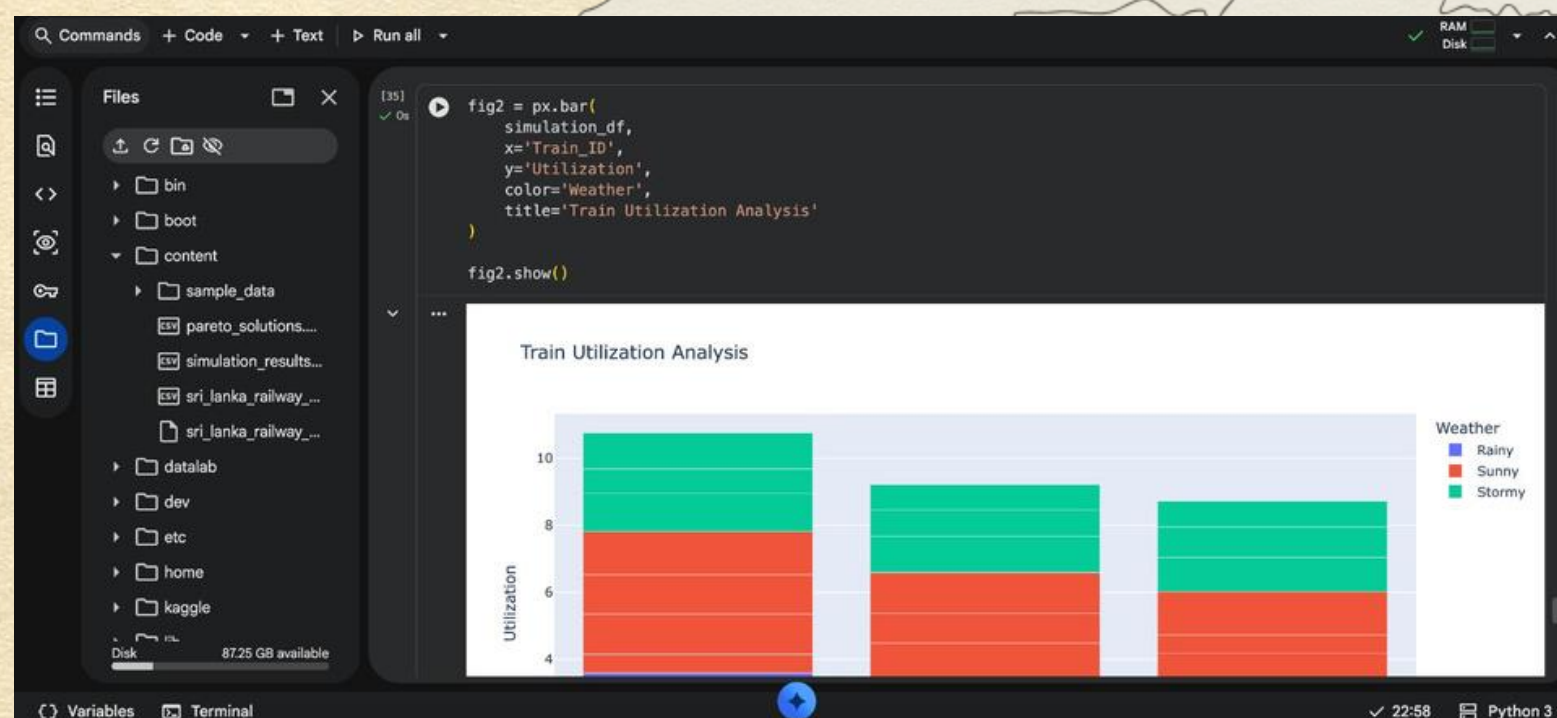
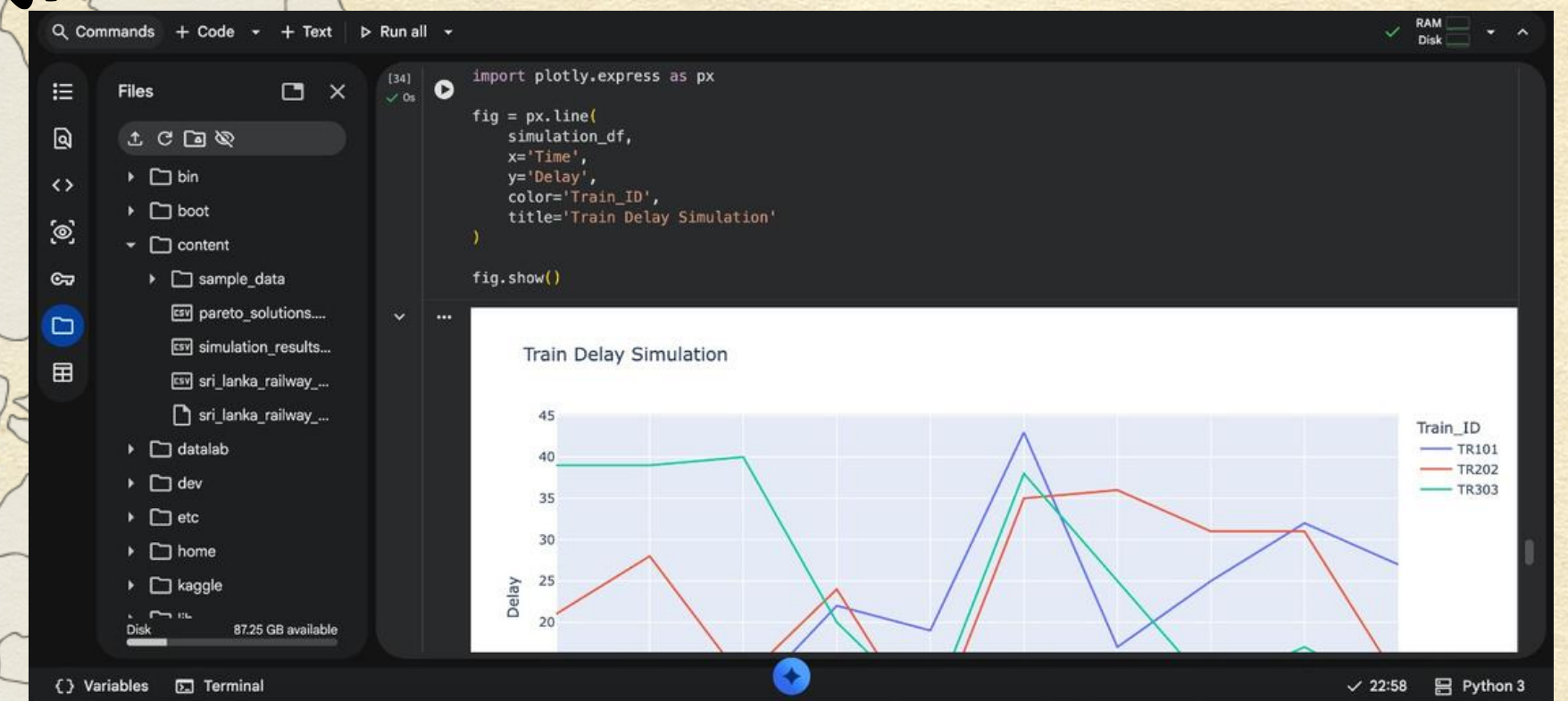
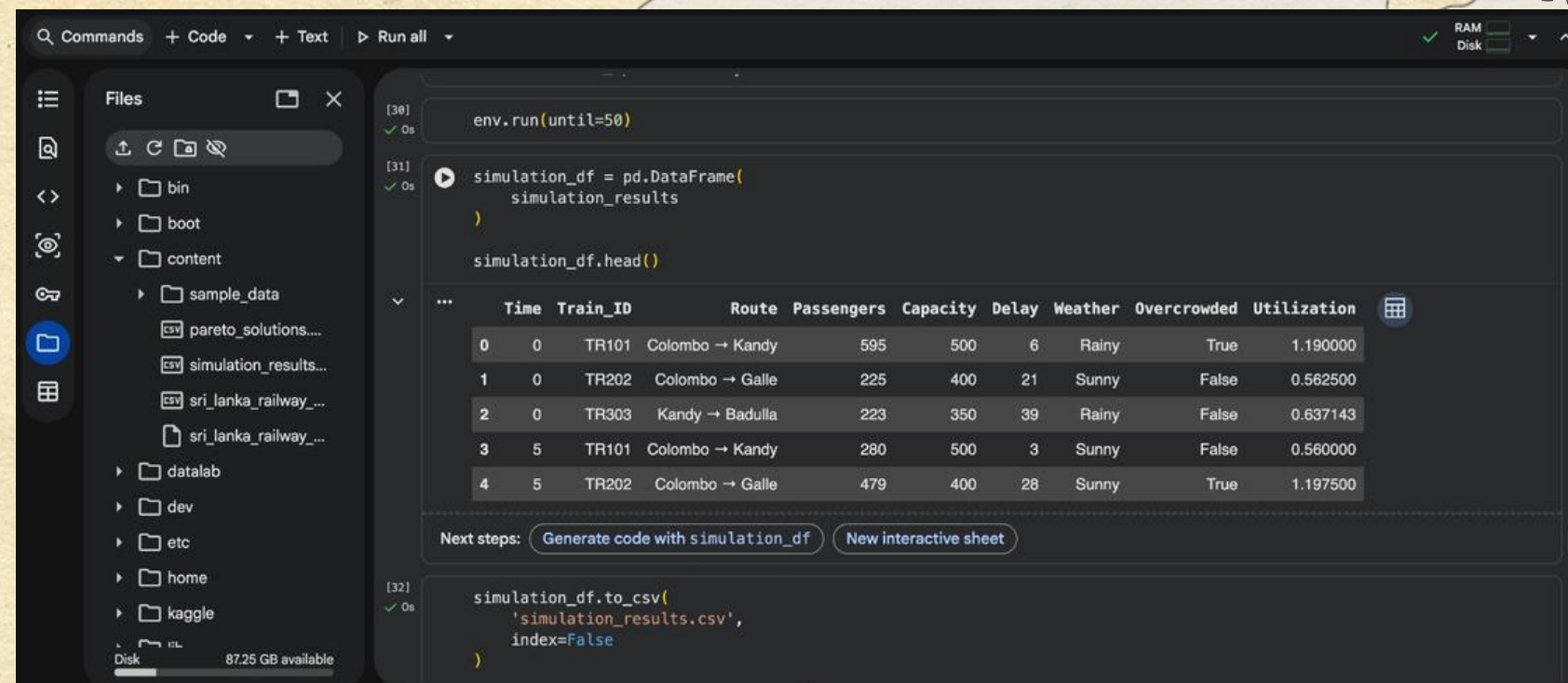
Time: 0
Train: TR202
Route: Colombo -> Galle

Passengers: 224
Capacity: 400

Delay: 30 mins
Weather: Rainy

Overcrowded: False
```

Prototype





Real-Time Train Tracking & Delay Prediction

IT22051202 - Didulantha C S

R26-IT-127

Real-Time Train Tracking & Delay Prediction

How the Solution Addresses the Problem:

- Lack of real-time train visibility
- Inability to anticipate delays
- Poor passenger decision-making

User Requirements Addressed

- Complete backend (Node.js + API routes)
- LSTM Python code for delay prediction
- Real-time train tracking
- Delay prediction display

⋮

Key Features:

- Alternative Boarding Stations
- Interactive Railway Map
- Multilingual + Theme Support



Real-Time Train Tracking & Delay Prediction

Design Excellence / Contribution:

Real-Time + Predictive Fusion

- Integrated live telemetry tracking with predictive delay analytics, moving the system beyond static schedule display into proactive journey intelligence.

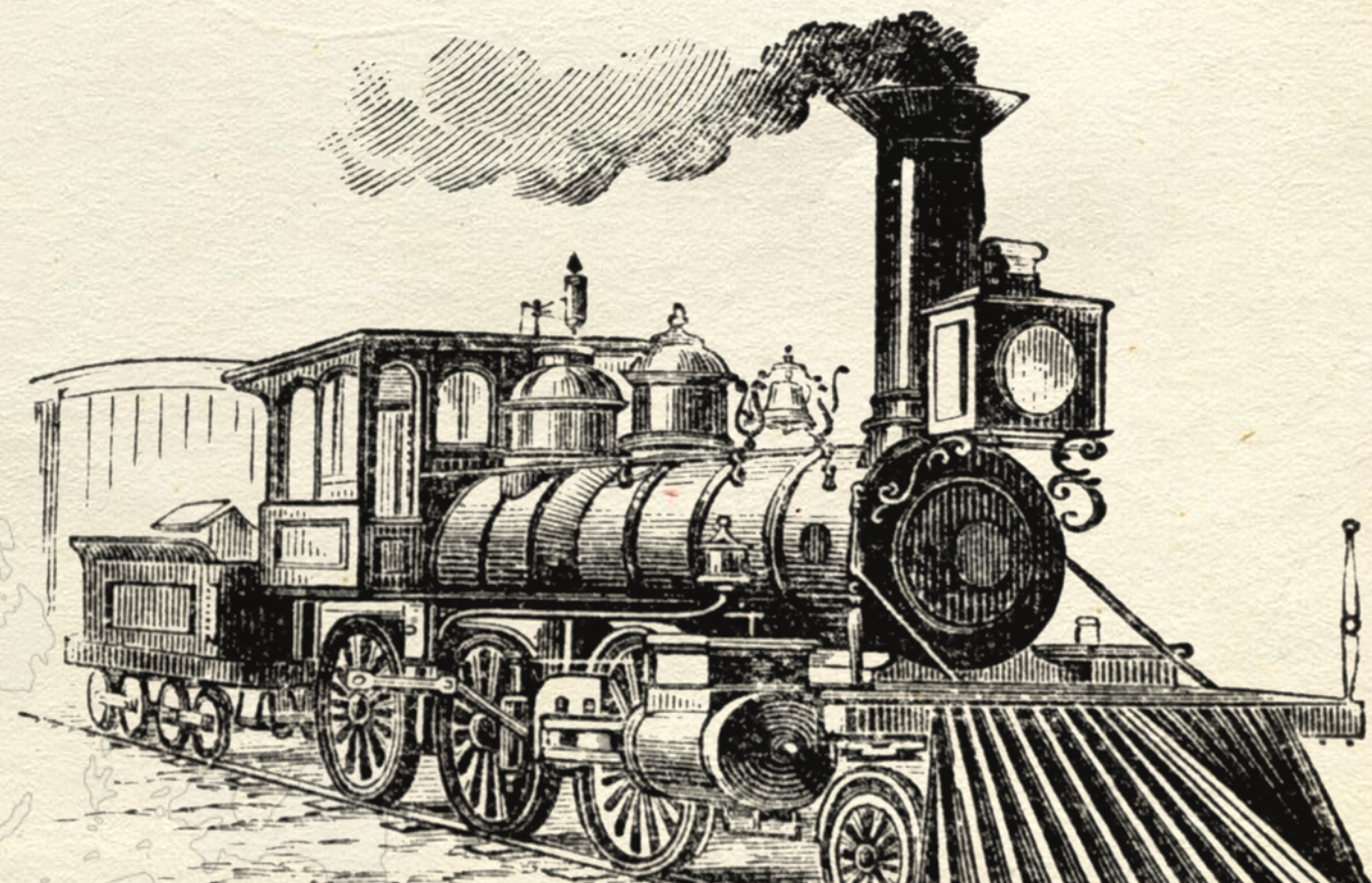
Map-Centric Interaction Quality:

Prototype Implementation:

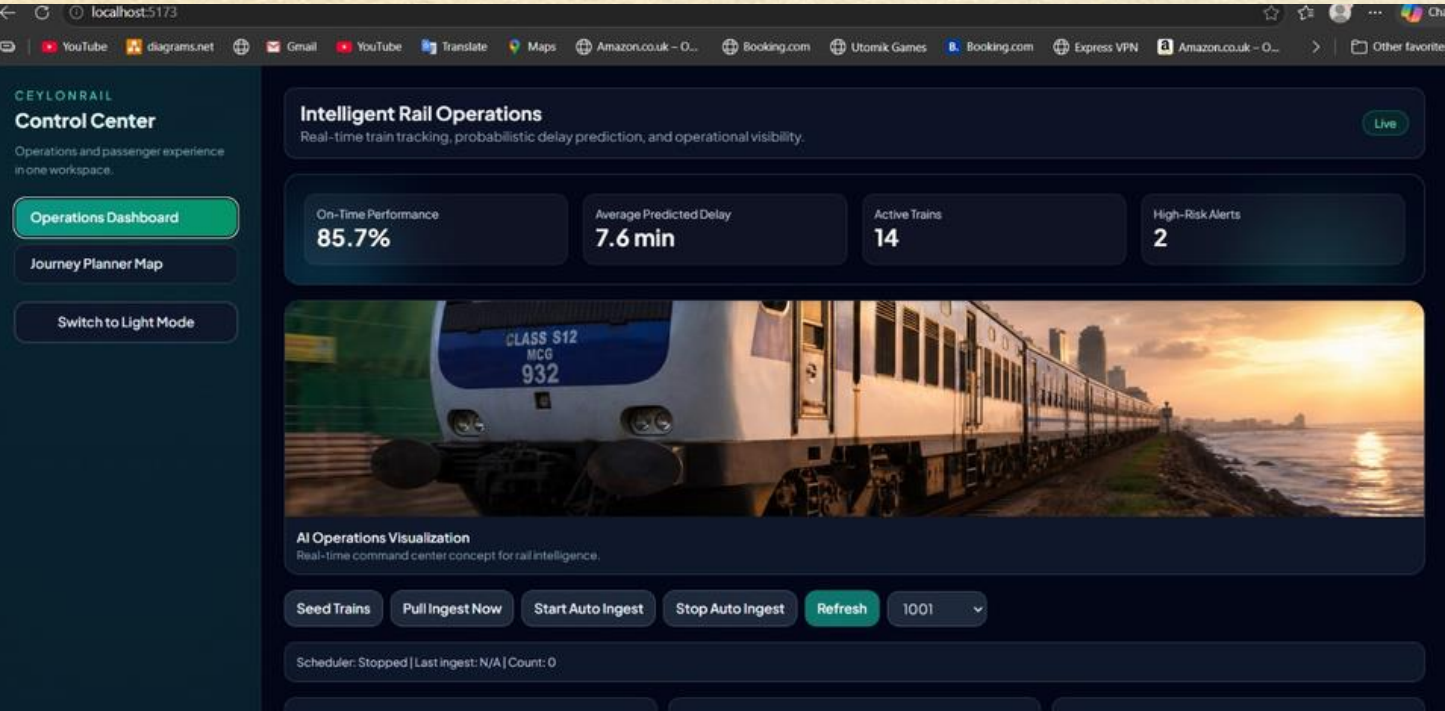
- Frontend Prototype (React + Tailwind + Leaflet)
- Backend Prototype (Node.js + Express + MongoDB)
- ML Prototype (Python Flask +

User Feedback

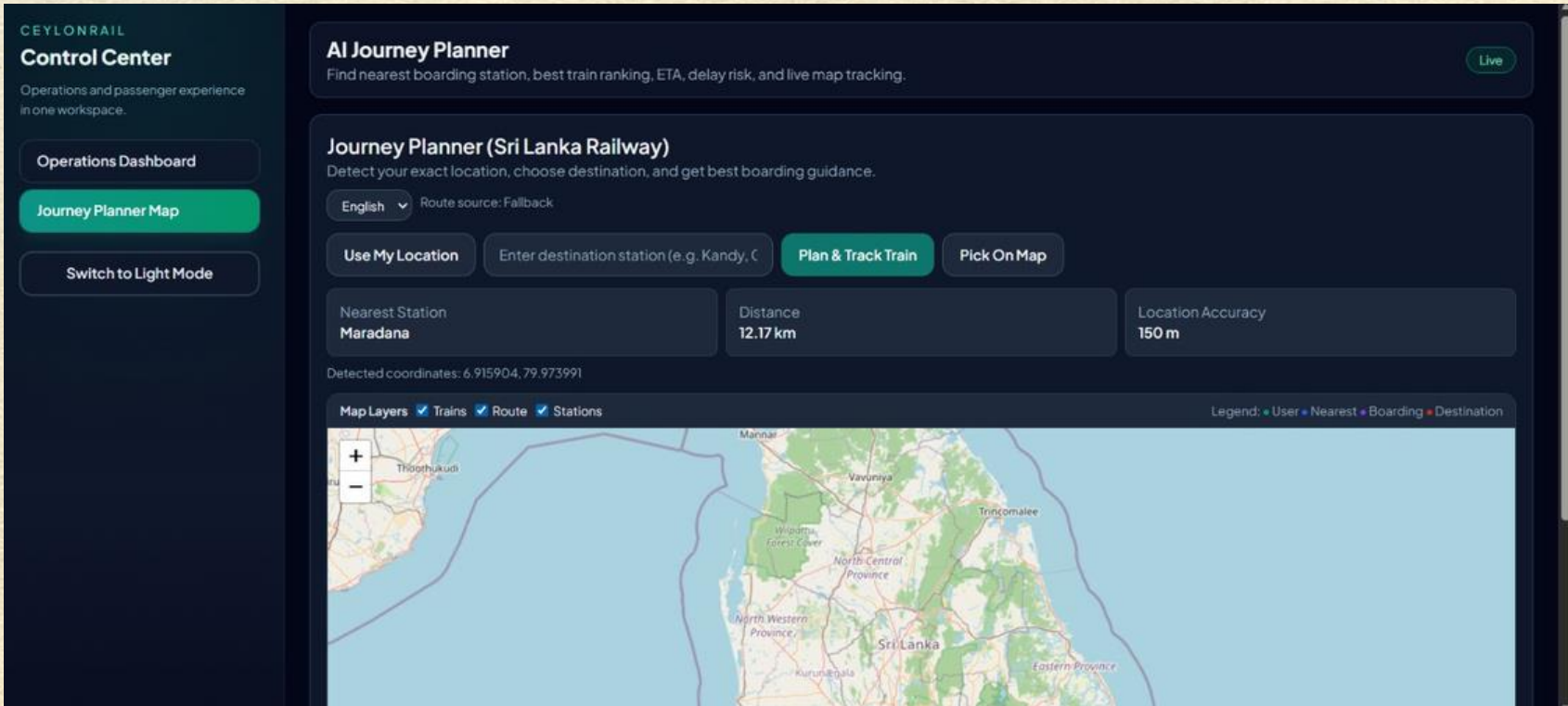
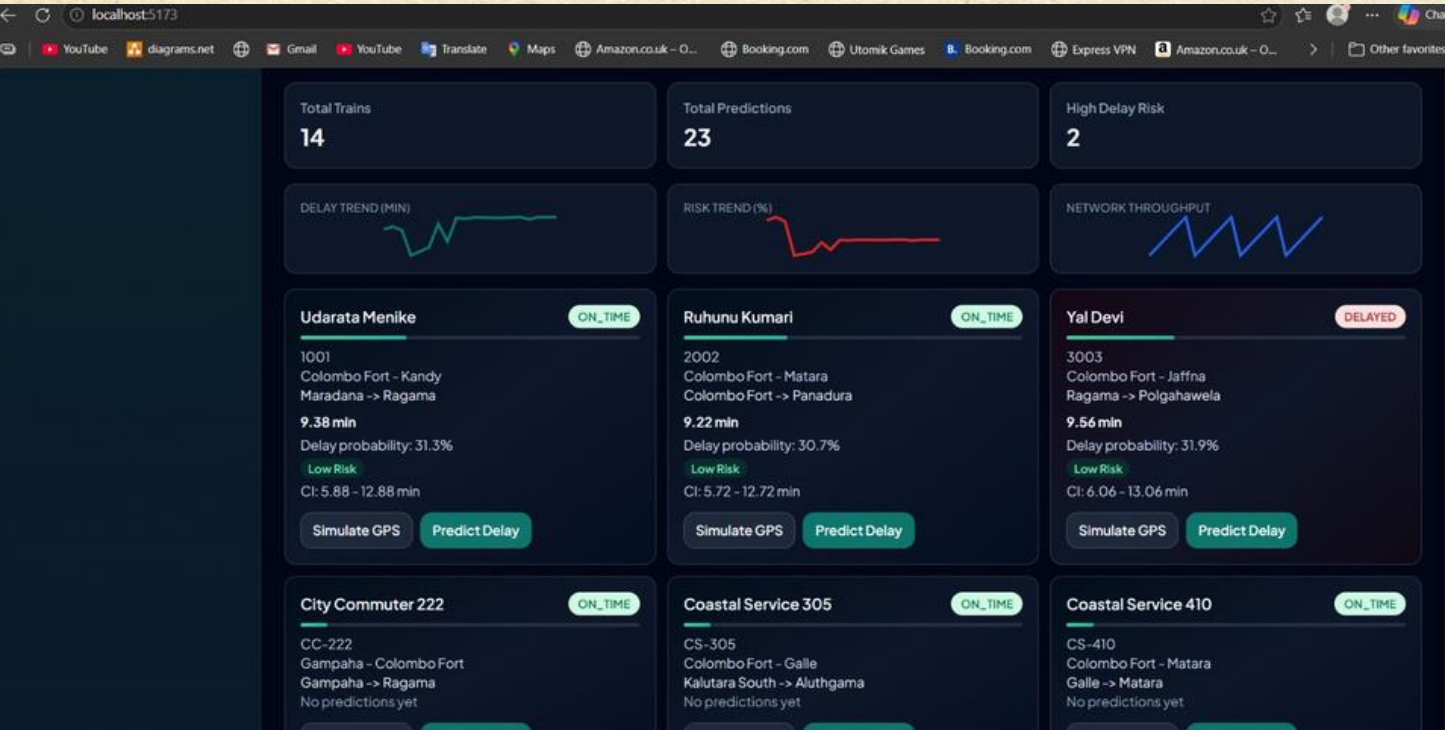
- Journey Planner is the highest-value user feature
- Accuracy expectations are high
- UI quality matters, but must stay



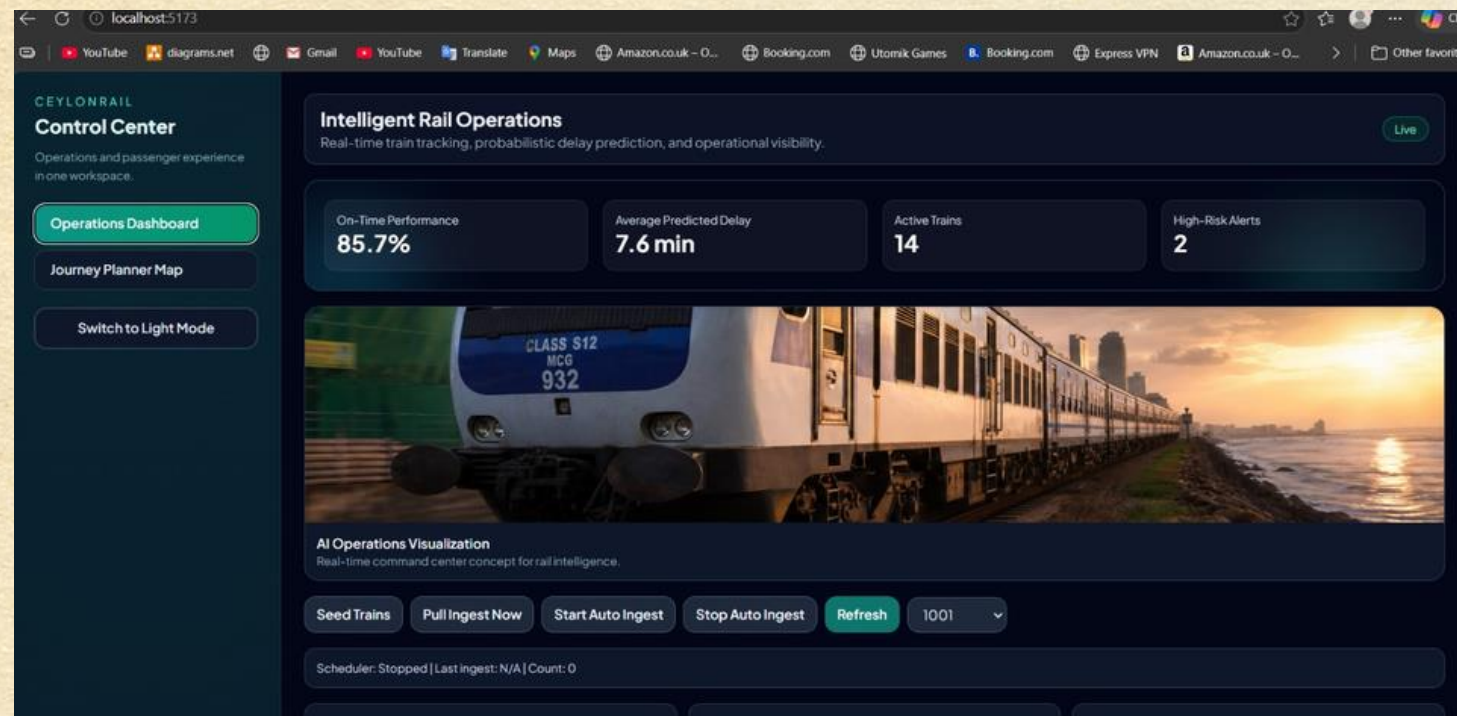
Prototype



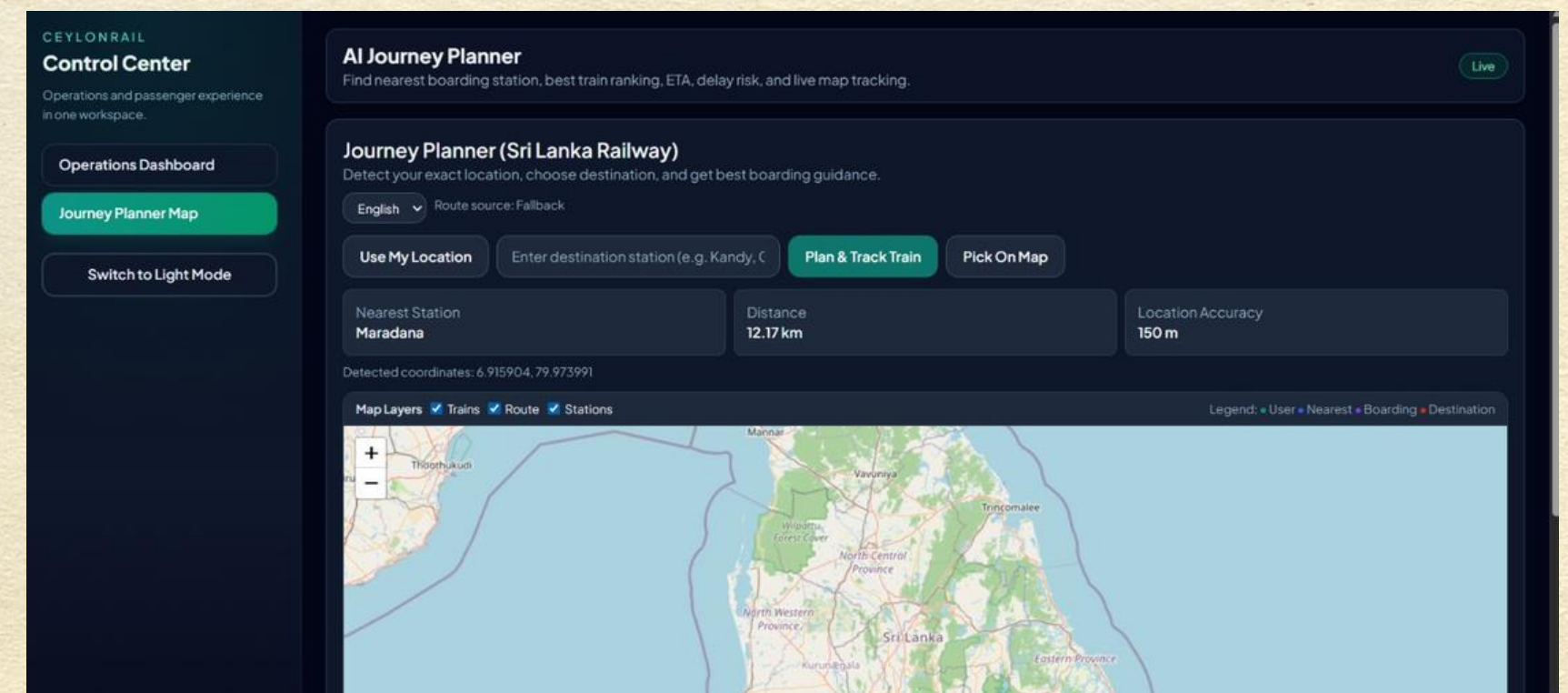
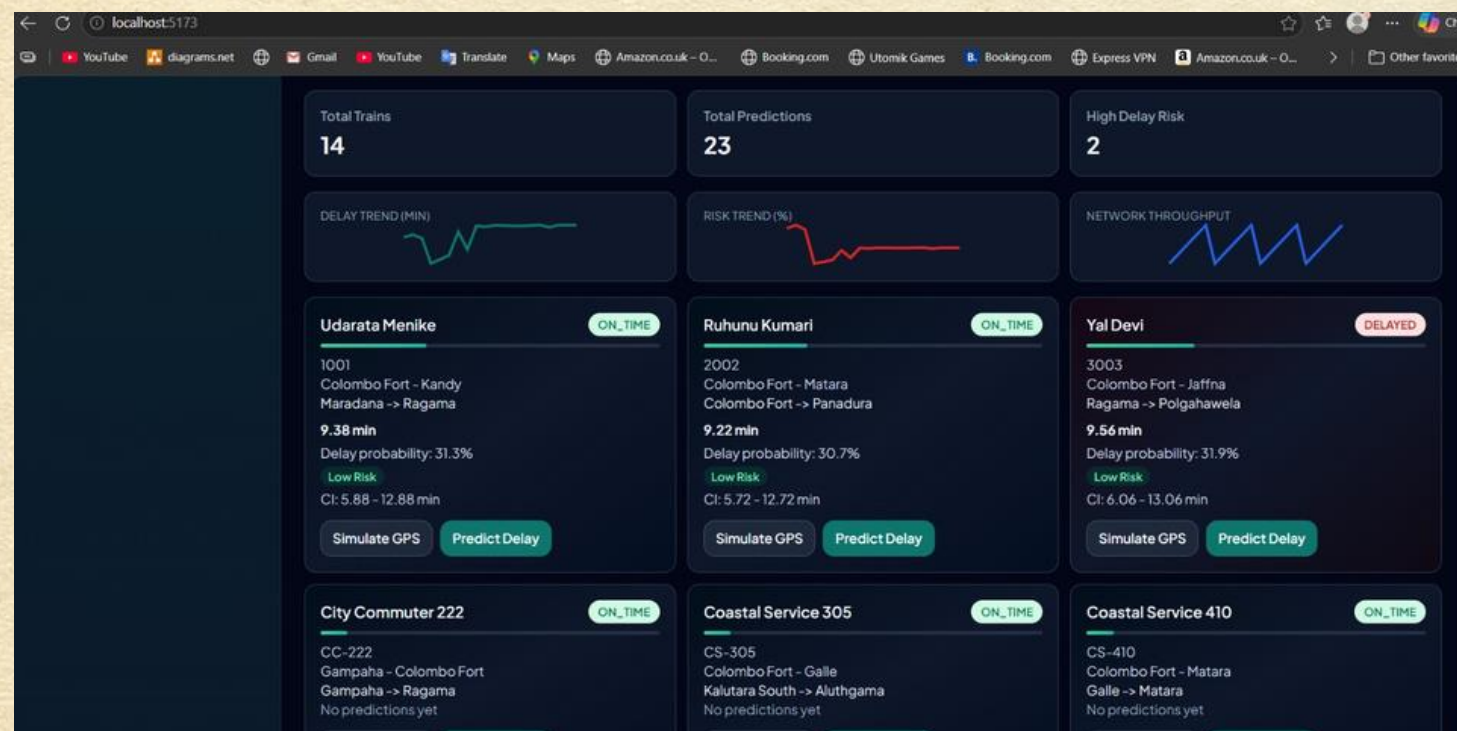
Recent Telemetry (3003)			
Time	Speed (km/h)	Observed Delay	Coordinates
5/9/2026, 9:02:34 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:33 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:29 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:58:30 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:11 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:10 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:53:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:52:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:51:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:50:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:49:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:48:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:47:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:46:24 PM	38.00	10.00	7.4800, 80.3300



Prototype



Recent Telemetry (3003)			
Time	Speed (km/h)	Observed Delay	Coordinates
5/9/2026, 9:02:34 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:33 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 9:02:29 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:58:30 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:12 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:11 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:54:10 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:53:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:52:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:51:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:50:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:49:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:48:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:47:24 PM	38.00	10.00	7.4800, 80.3300
5/9/2026, 8:46:24 PM	38.00	10.00	7.4800, 80.3300



Prototype

Operations Dashboard

Journey Planner Map

Switch to Light Mode

Journey Planner (Sri Lanka Railway)

Detect your exact location, choose destination, and get best boarding guidance.

English

Route source: FALLBACK_STATION_ORDER

Use My Location

kandy

Plan & Track Train

Pick On Map

Nearest Station
Maradana

Distance
12.17 km

Location Accuracy
150 m

Detected coordinates: 6.915904, 79.973991

Recommended Boarding: Colombo Fort ? Kandy

Exact arrival at boarding station: 2:41:04 AM | Delay: 0.7 min | ML risk: 31.4%

Best Train Ranking

HS-808 - Hill Service 808

Arrive boarding: 2:41:04 AM | Delay: 0.7 min | Trip: 211.4 min | Score: 33.45

ML Delay Risk: 31.4% | ML Predicted Delay: 9.4 min | CI: 5.9 - 12.9

1001 - Udarata Menike

Arrive boarding: 2:52:27 AM | Delay: 4 min | Trip: 136.9 min | Score: 31.87

ML Delay Risk: 31.3% | ML Predicted Delay: 9.4 min | CI: 5.9 - 12.9

IC-101 - Intercity Express 101

Arrive boarding: 2:41:50 AM | Delay: 21.5 min | Trip: 68.7 min | Score: 26.47

ML Delay Risk: 31.8% | ML Predicted Delay: 9.6 min | CI: 6.0 - 13.1

MC-520 - Mainline Commuter 520

Arrive boarding: 2:47:18 AM | Delay: 21.6 min | Trip: 132.6 min | Score: 3.55

ML Delay Risk: 32.5% | ML Predicted Delay: 9.8 min | CI: 6.3 - 13.3

R26-IT-127

IT22051202 - Didulantha C S

AI-Based Railway Fraud Detection & Passenger Verification System

Machine Learning-based detection of suspicious railway ticket transactions

IT22126610 – Muthukudaarachchi V.U

R26-IT-127



AI-Based Railway Fraud Detection System

How the Solution Addresses the Problem:

- Detects suspicious ticket transactions using ML
- Reduces manual inspection errors
- Identifies high-risk passengers in real-time

User Requirements Addressed:

- Real-time fraud detection
- Interactive dashboard
- Risk level classification (Low / Medium / High)
- Filtering (Train ID, Route, Time)

Key Features:

- Anomaly detection using ML
- Risk scoring system
- Live prediction interface



AI-Based Railway Fraud Detection System

Design Excellence / Contribution:

Implemented 3 ML models:

- Isolation Forest
- One-Class SVM
- Autoencoder

Compared using:

- Accuracy, Precision, Recall, F1 Score

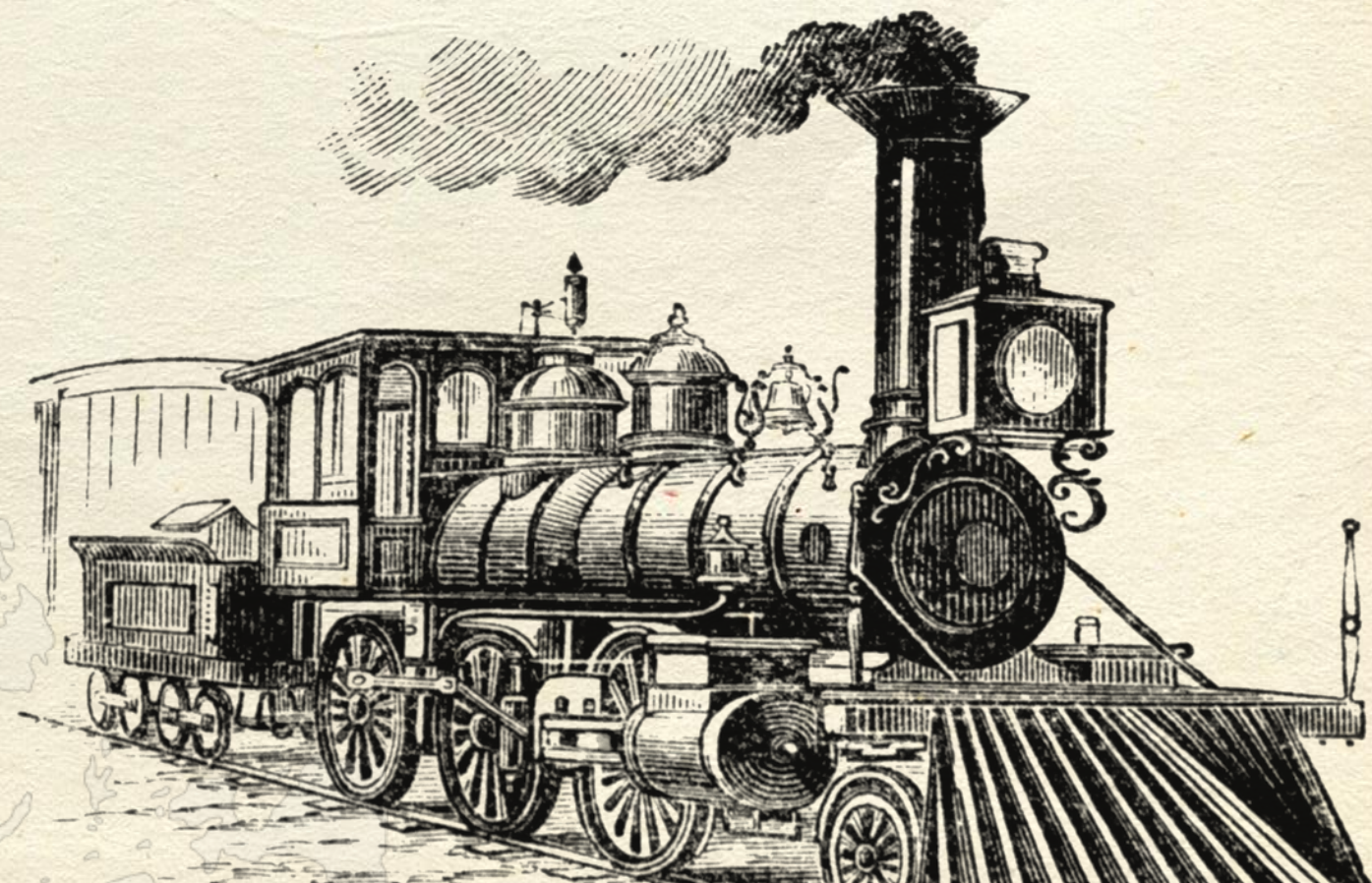
Model	Accuracy	Precision	Recall	F1 Score
Isolation Forest	9.414%	1.421%	3.122%	1.953%
One-Class SVM	9.441%	1.637%	3.539%	★ 22.39%
Autoencoder	9.363%	909%	1.998%	1.250%

Prototype Implementation:

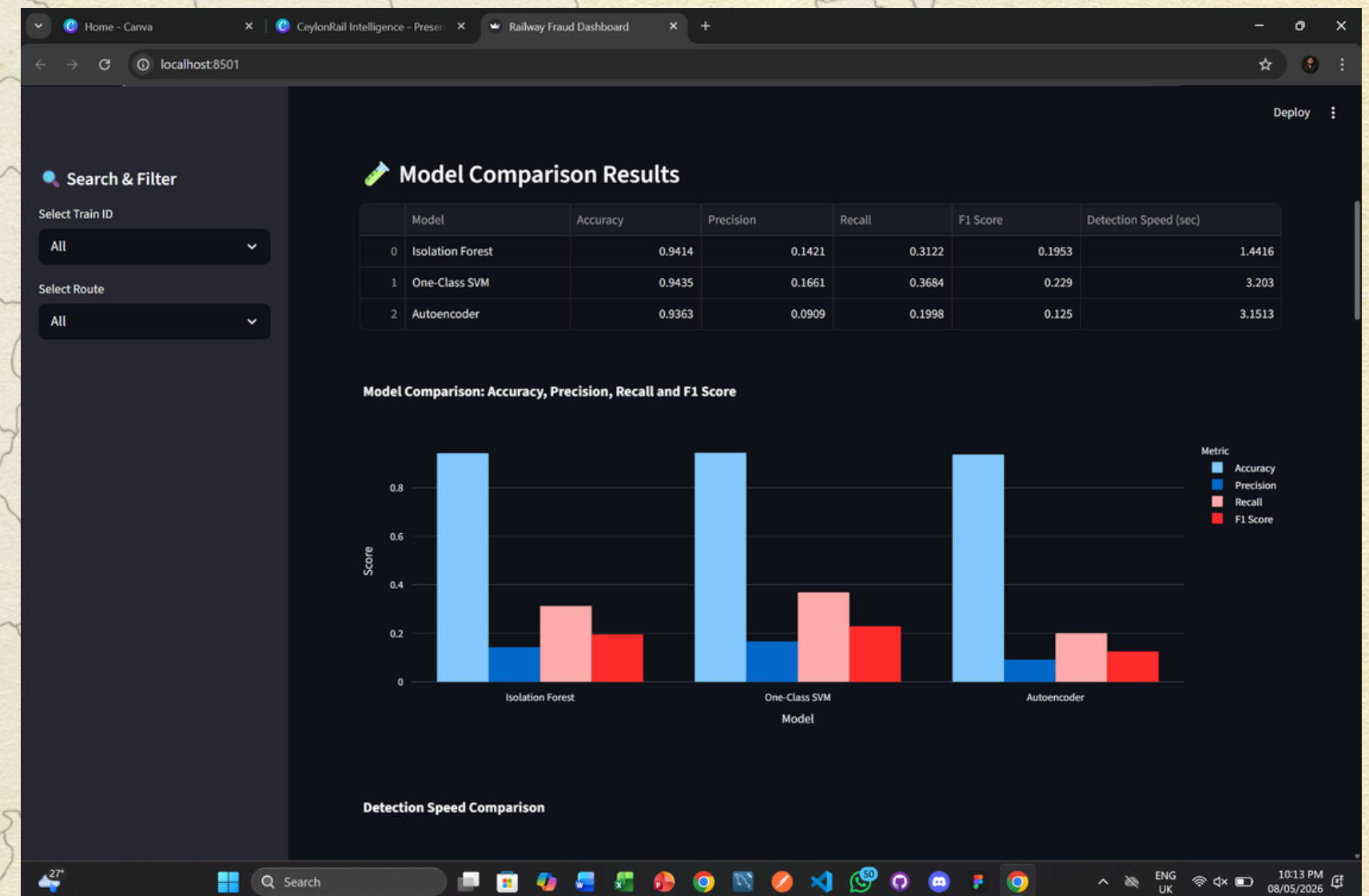
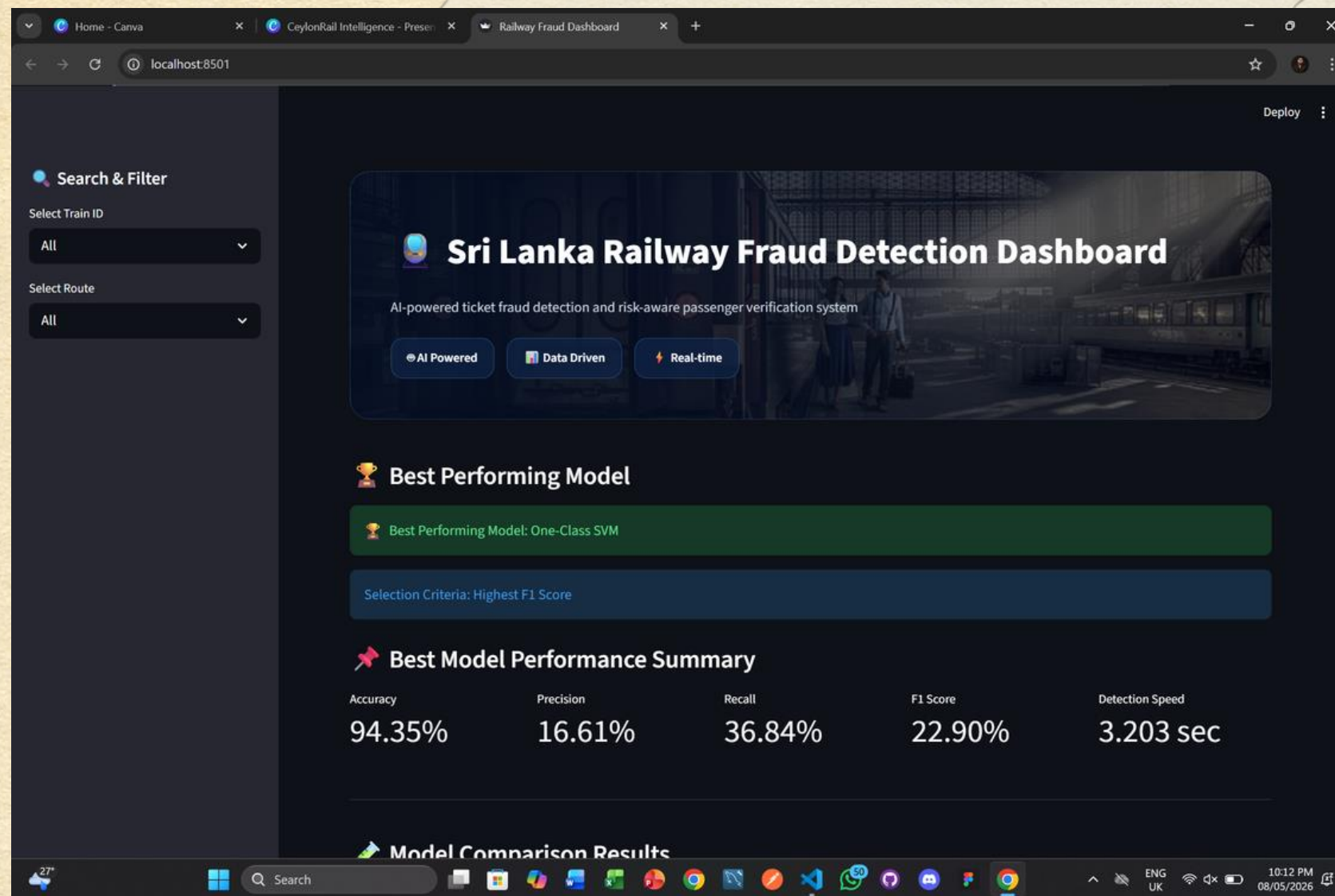
- Real-time fraud detection system
- Interactive dashboard
- Live prediction feature

User Feedback

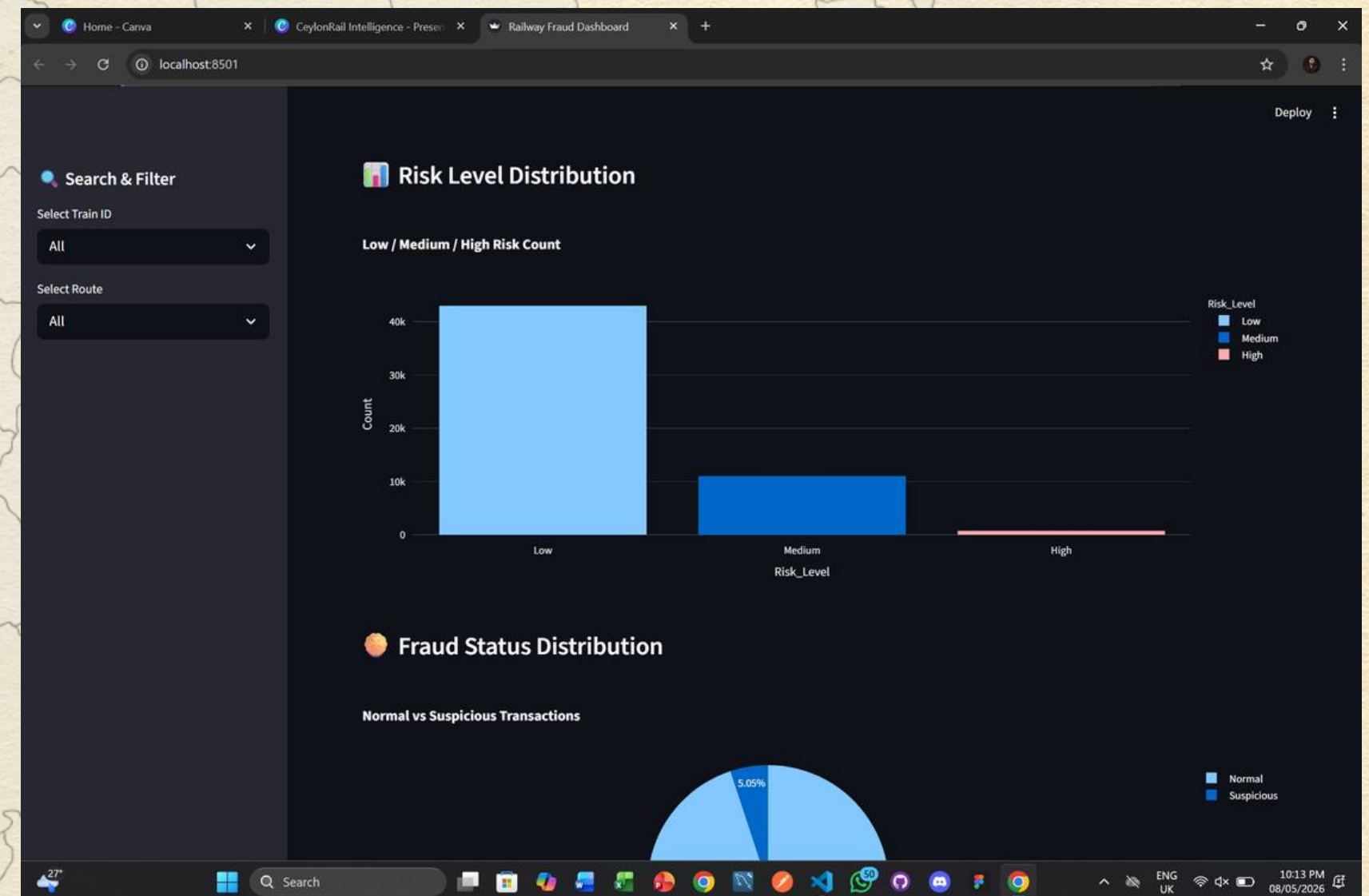
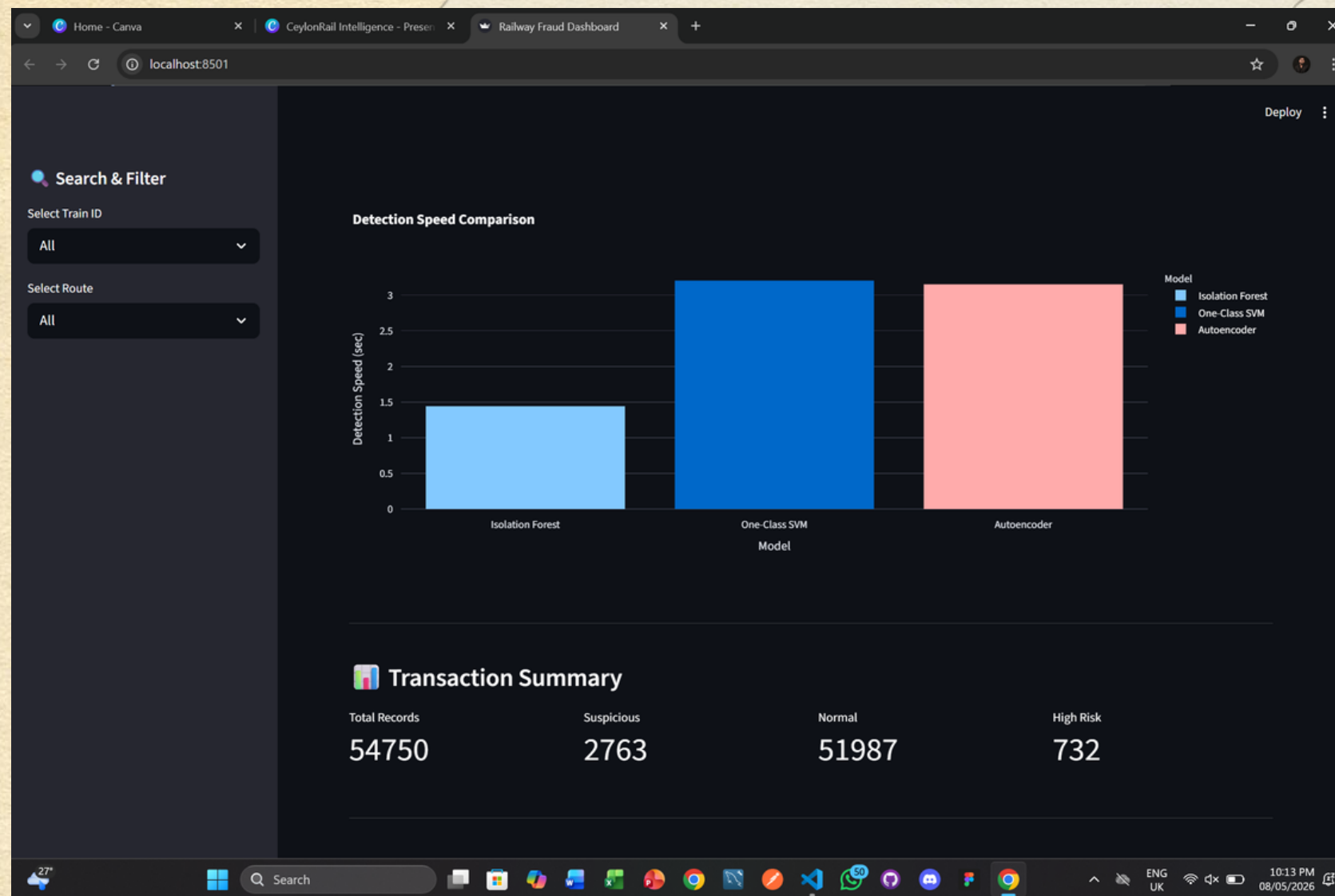
- Easy to use and understand
- Clear visualization of fraud patterns
- Useful for decision making



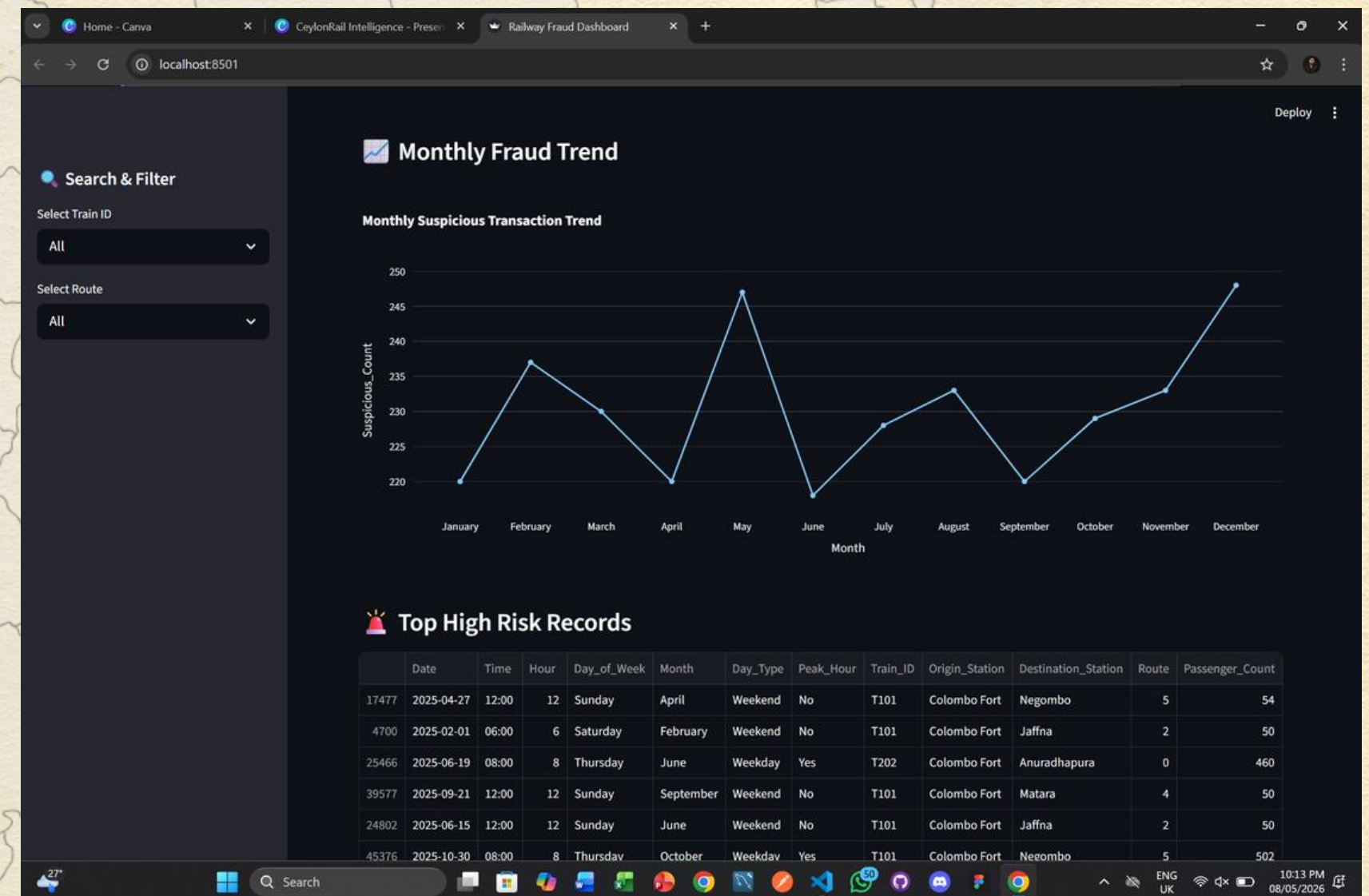
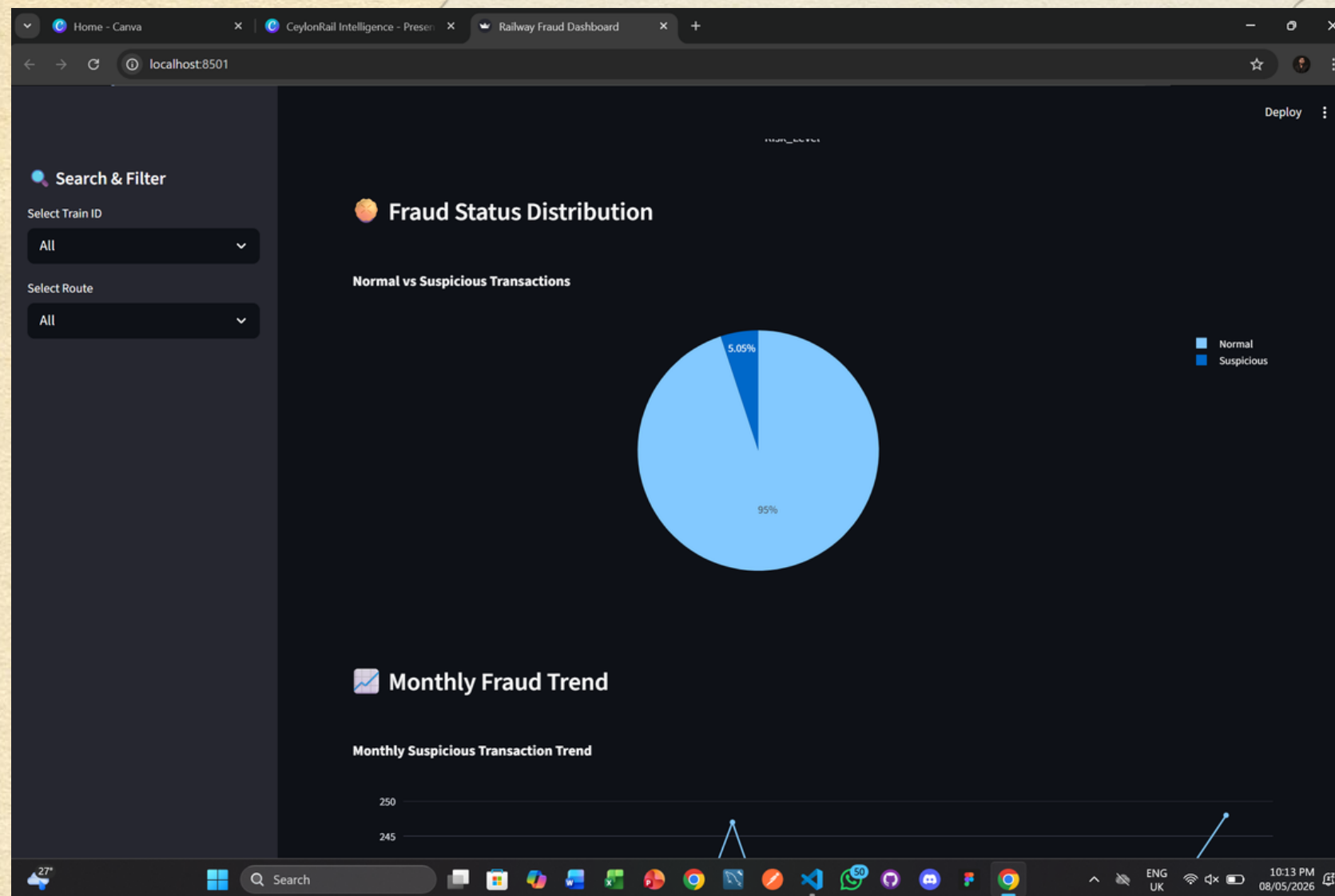
Prototype



Prototype



Prototype



Prototype

Search & Filter

Select Train ID

All

Select Route

All

🔔 Top High Risk Records

	Date	Time	Hour	Day_of_Week	Month	Day_Type	Peak_Hour	Train_ID	Origin_Station	Destination_Station	Route	Passenger_Count
17477	2025-04-27	12:00	12	Sunday	April	Weekend	No	T101	Colombo Fort	Negombo	5	54
4700	2025-02-01	06:00	6	Saturday	February	Weekend	No	T101	Colombo Fort	Jaffna	2	50
25466	2025-06-19	08:00	8	Thursday	June	Weekday	Yes	T202	Colombo Fort	Anuradhapura	0	460
39577	2025-09-21	12:00	12	Sunday	September	Weekend	No	T101	Colombo Fort	Matara	4	50
24802	2025-06-15	12:00	12	Sunday	June	Weekend	No	T101	Colombo Fort	Jaffna	2	50
45376	2025-10-30	08:00	8	Thursday	October	Weekday	Yes	T101	Colombo Fort	Negombo	5	502
35332	2025-08-24	12:00	12	Sunday	August	Weekend	No	T102	Colombo Fort	Negombo	5	50
22602	2025-05-31	12:00	12	Saturday	May	Weekend	No	T101	Colombo Fort	Anuradhapura	0	50
39542	2025-09-21	12:00	12	Sunday	September	Weekend	No	T202	Colombo Fort	Negombo	5	52
21449	2025-05-23	19:00	19	Friday	May	Weekday	Yes	T301	Colombo Fort	Matara	4	452

Download Suspicious Records CSV

📄 Full Filtered Result Data

	Date	Time	Hour	Day_of_Week	Month	Day_Type	Peak_Hour	Train_ID	Origin_Station	Destination_Station	Route	Passenger_Count
0	2025-01-01	06:00	6	Wednesday	January	Weekday	No	T101	Colombo Fort	Kandy	3	296
1	2025-01-01	08:00	8	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	391
2	2025-01-01	12:00	12	Wednesday	January	Weekday	No	T101	Colombo Fort	Kandy	3	134
3	2025-01-01	17:00	17	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	332
4	2025-01-01	19:00	19	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	369

Search & Filter

Select Train ID

All

Select Route

All

📄 Full Filtered Result Data

	Date	Time	Hour	Day_of_Week	Month	Day_Type	Peak_Hour	Train_ID	Origin_Station	Destination_Station	Route	Passenger_Count
0	2025-01-01	06:00	6	Wednesday	January	Weekday	No	T101	Colombo Fort	Kandy	3	296
1	2025-01-01	08:00	8	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	391
2	2025-01-01	12:00	12	Wednesday	January	Weekday	No	T101	Colombo Fort	Kandy	3	134
3	2025-01-01	17:00	17	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	332
4	2025-01-01	19:00	19	Wednesday	January	Weekday	Yes	T101	Colombo Fort	Kandy	3	369
5	2025-01-01	06:00	6	Wednesday	January	Weekday	No	T102	Colombo Fort	Kandy	3	115
6	2025-01-01	08:00	8	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	298
7	2025-01-01	12:00	12	Wednesday	January	Weekday	No	T102	Colombo Fort	Kandy	3	106
8	2025-01-01	17:00	17	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	439
9	2025-01-01	19:00	19	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	360

🔒 Live Fraud Risk Prediction

Enter new passenger transaction details and predict fraud risk instantly.

Hour

8

Temperature

30

Ticket Type for Prediction

0

Passenger Count

250

Rainfall

0

Class Type for Prediction

0

Prototype

Search & Filter

Select Train ID

All

Select Route

All

Deploy

6	2025-01-01	08:00	8	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	298
7	2025-01-01	12:00	12	Wednesday	January	Weekday	No	T102	Colombo Fort	Kandy	3	106
8	2025-01-01	17:00	17	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	439
9	2025-01-01	19:00	19	Wednesday	January	Weekday	Yes	T102	Colombo Fort	Kandy	3	360

Live Fraud Risk Prediction

Enter new passenger transaction details and predict fraud risk instantly.

Hour

8

Temperature

30

Ticket Type for Prediction

0

Passenger Count

250

Rainfall

0

Class Type for Prediction

0

Available Seats

150

Route for Prediction

0

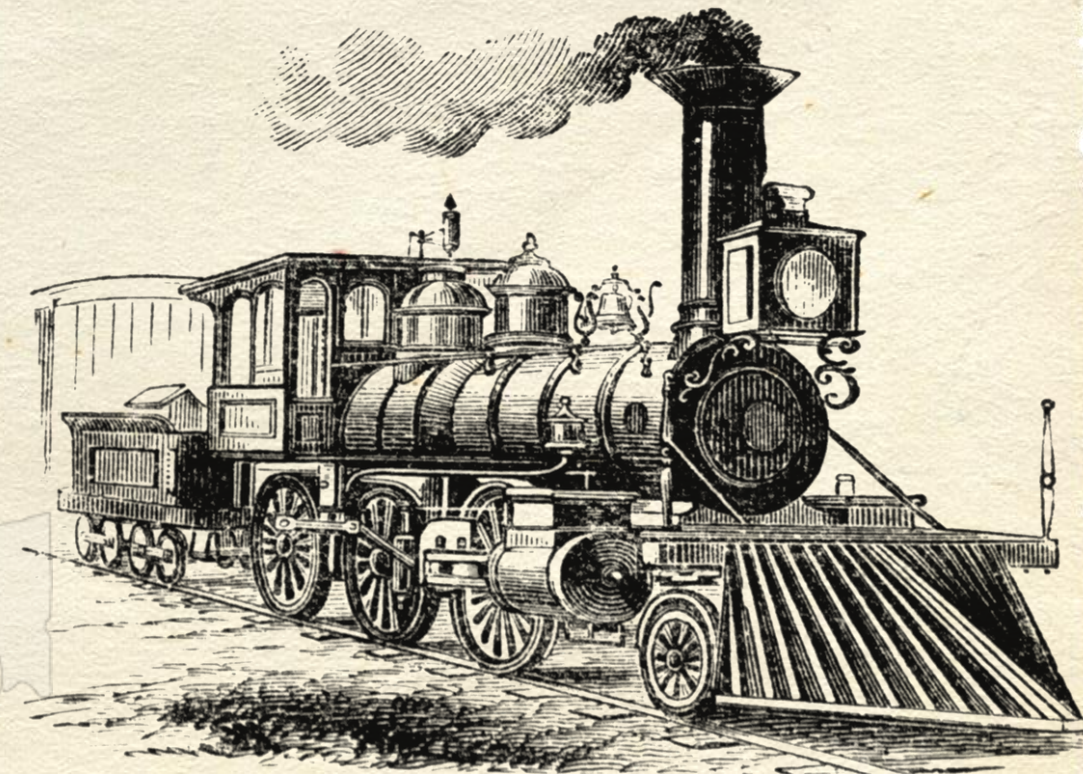
Predict Fraud Risk



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI

IT 22151506 - Senawirathna D.M.N.T

R26-IT-127



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI

User Requirements Addressed:

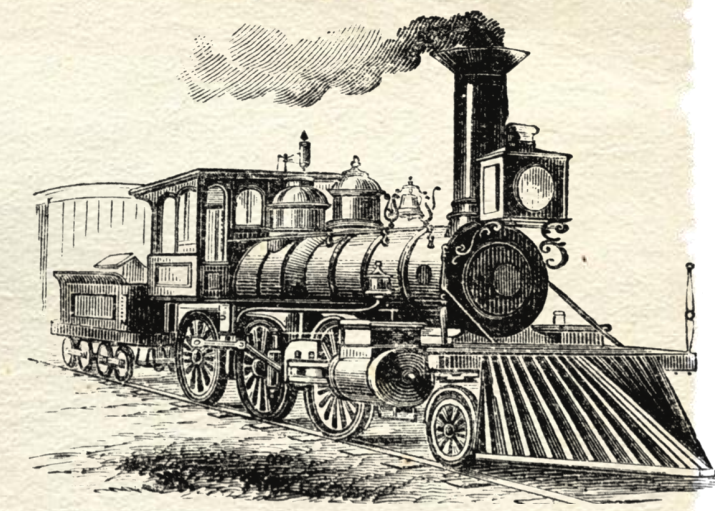
How Solution Addresses Sub-Problem:

- Forecasts future railway passenger demand using AI
- Identifies high-demand routes and peak travel periods
- Supports intelligent railway capacity planning
- Helps reduce overcrowding in trains

Requirement		Solution	
	Reduce overcrowding		Demand forecasting
	Better seat management		Adaptive allocation
	Improve planning		AI analytics
	Better passenger experience		Predictive optimization



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI



Design Excellence:

- AI-driven passenger demand forecasting
- Adaptive seat allocation mechanism
- Sri Lanka-specific railway analytics
- Spatio-temporal demand analysis

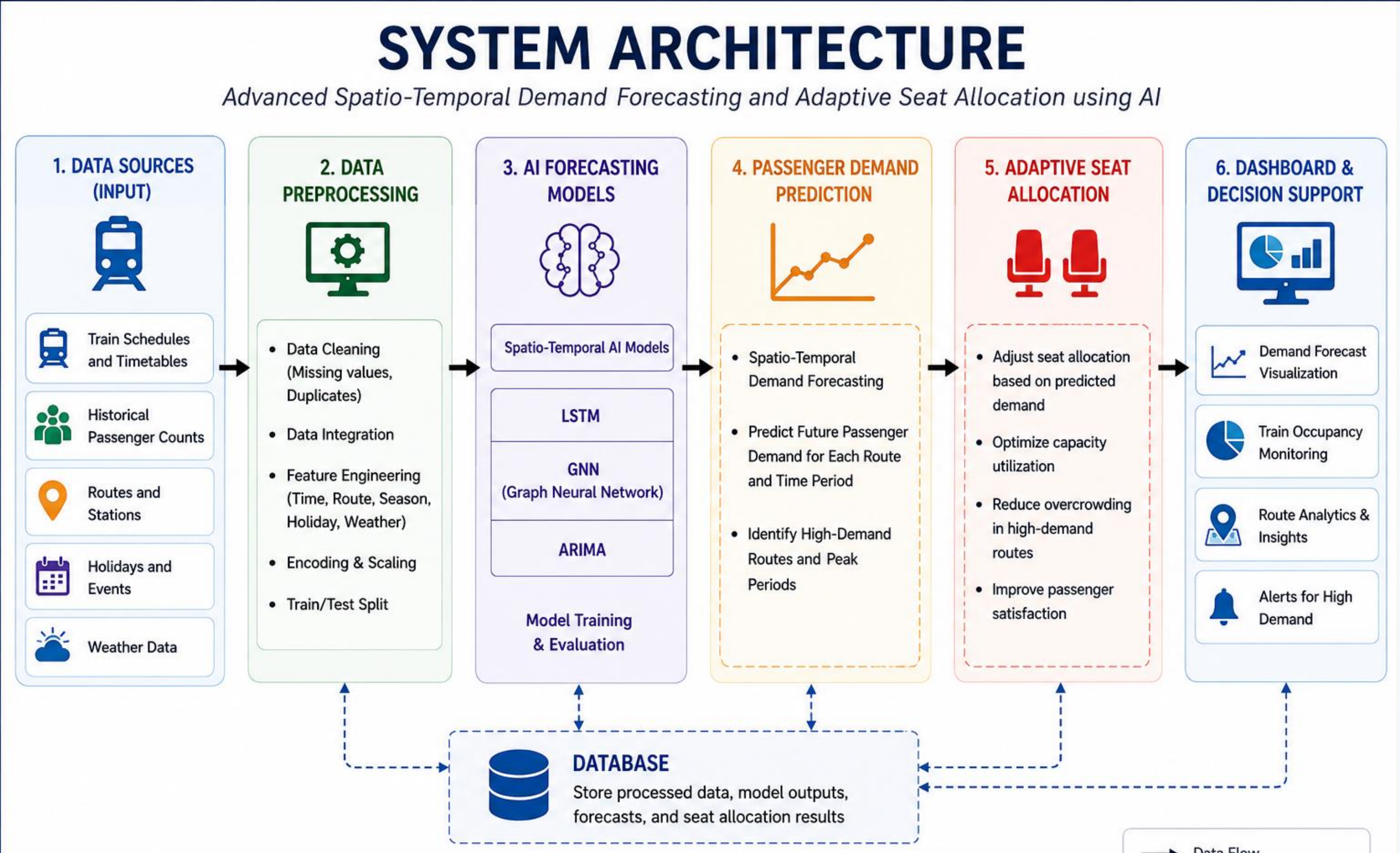
Contributions:

- Improves railway planning
- Reduces overcrowding
- Enhances seat utilization
- Supports intelligent decision-making

Expected User Feedback:

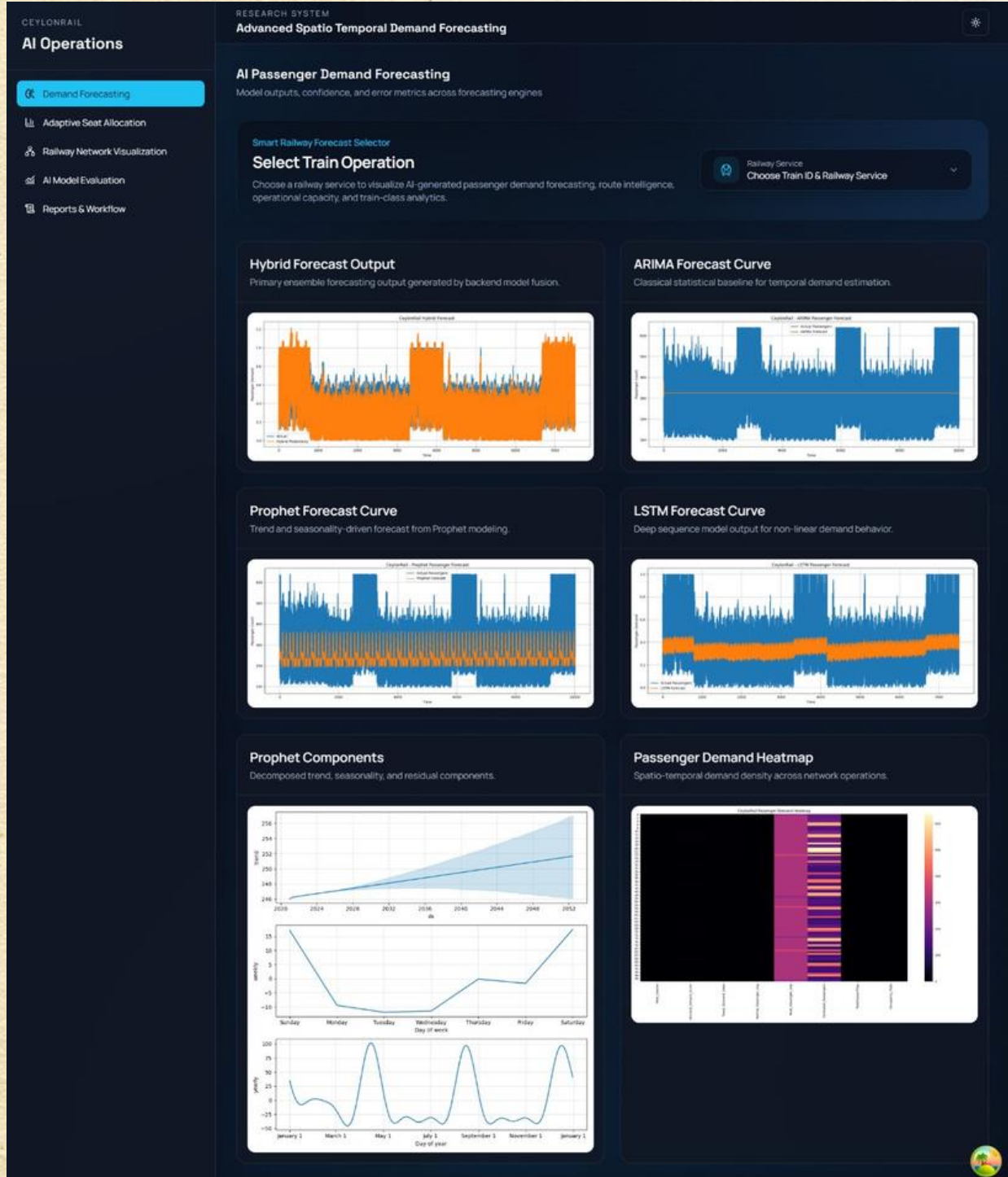
- Helps identify high-demand railway routes
- Supports better railway planning decisions
- Can improve seat allocation efficiency
- Expected to reduce overcrowding during peak periods

Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI



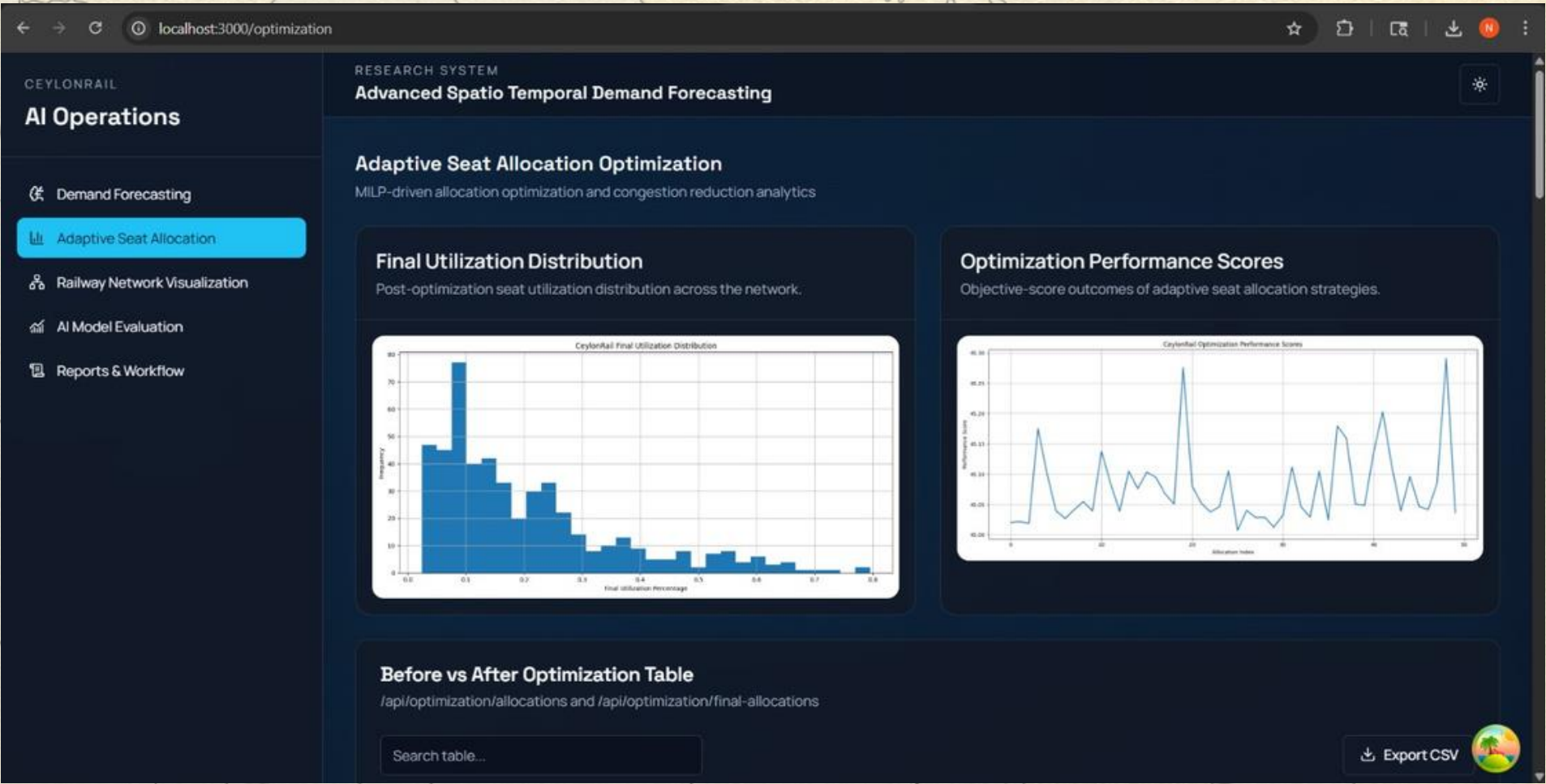
Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI

AI-Based Passenger Demand
Forecasting



Prototype

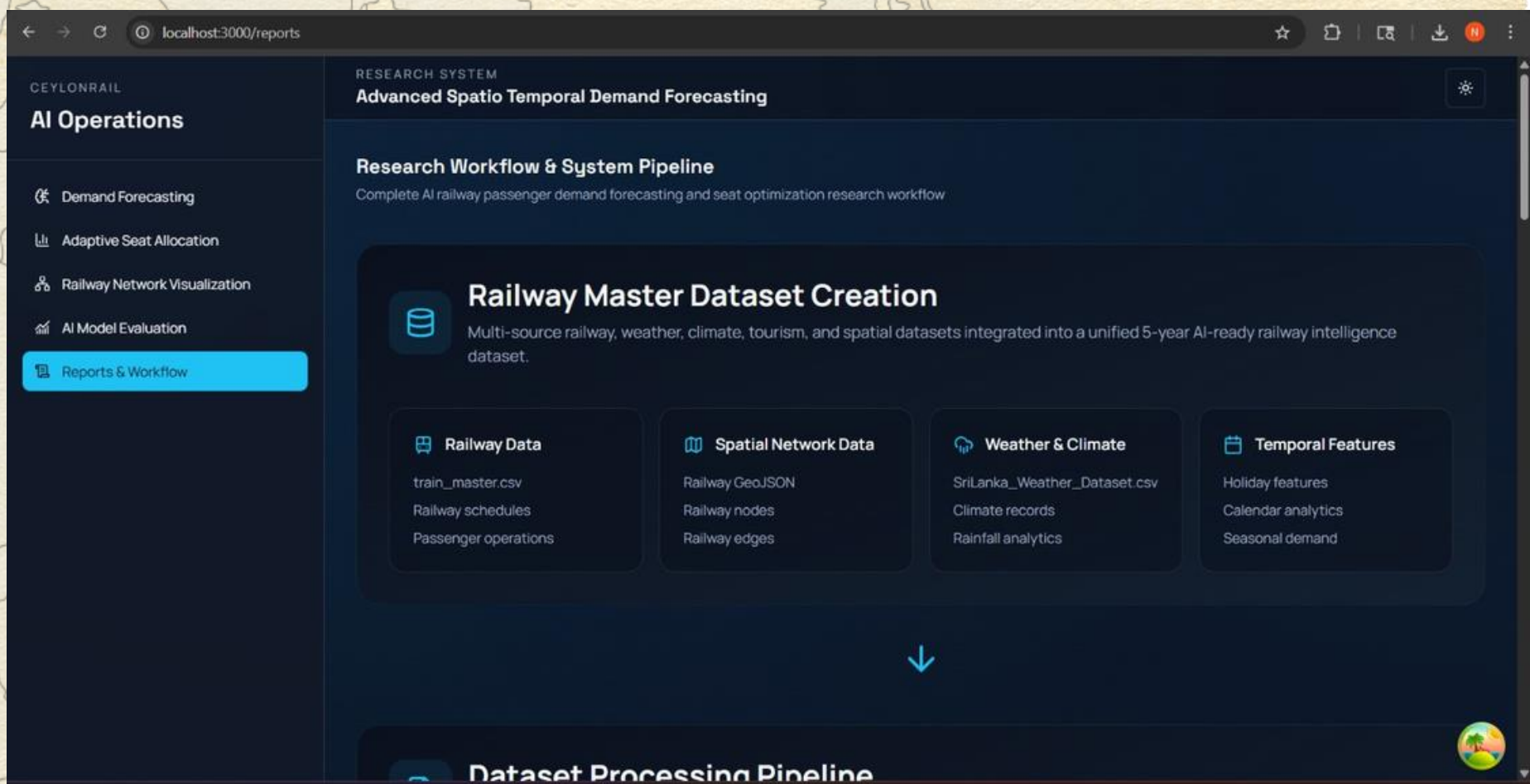
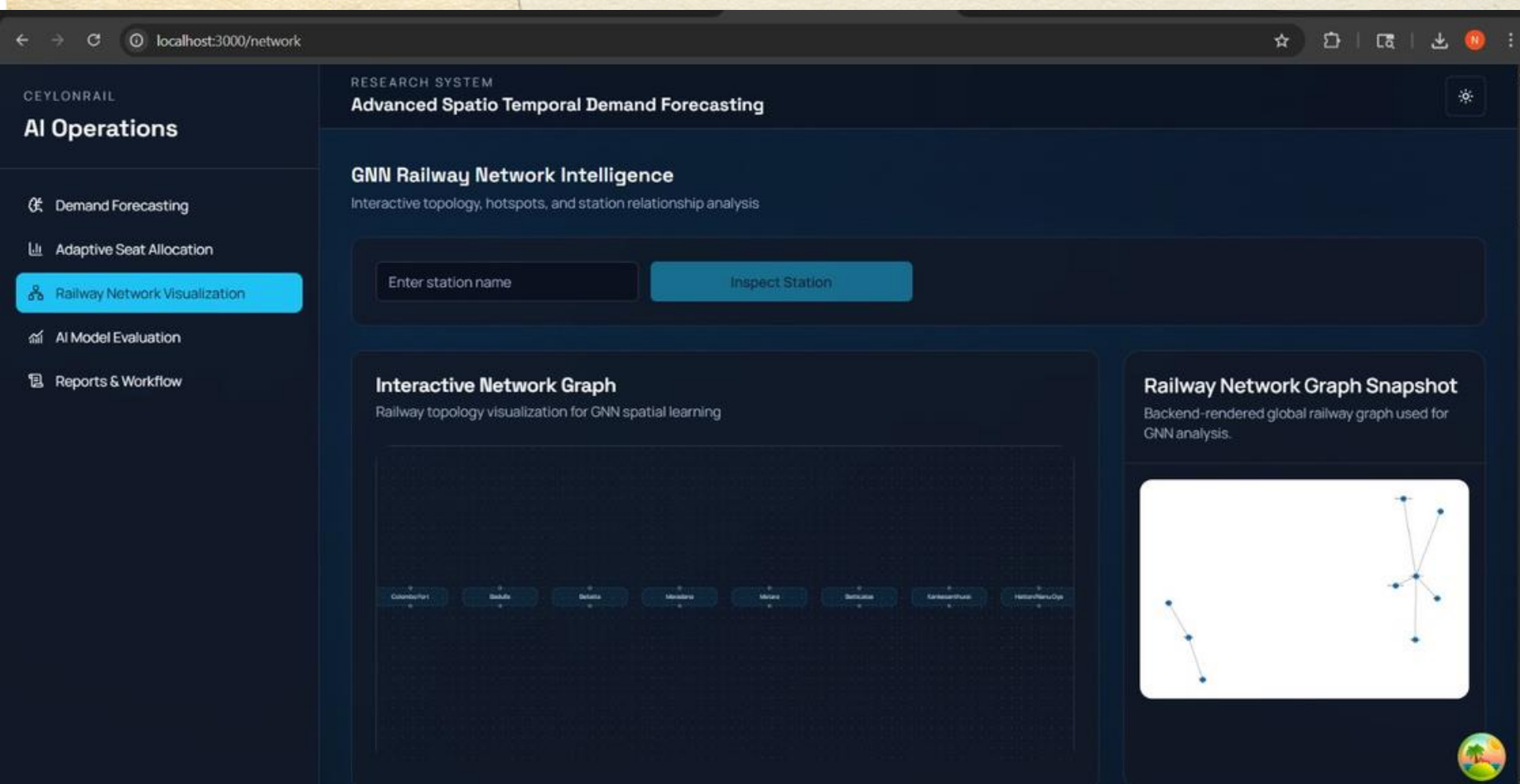
Adaptive Seat Allocation
Optimization



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI Prototype

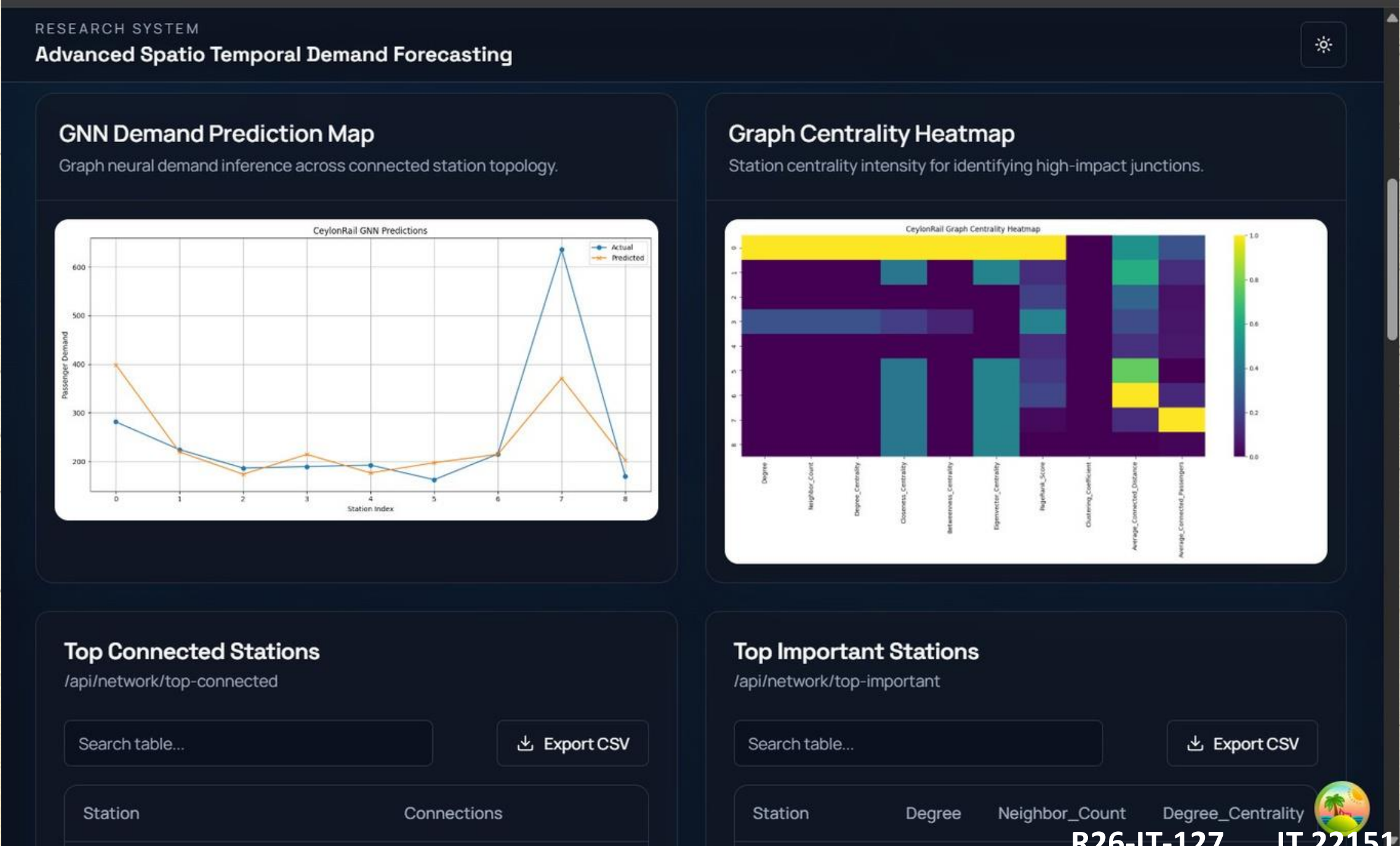
AI Model Performance Evaluation

Research Workflow & System Pipeline



Advanced Spatio-Temporal Demand Forecasting and Adaptive Seat Allocation using AI

Prototype Railway Network Visualization (GNN)



Current Progress

Component	Current PP1
Passenger Demand Forecasting	Dataset prep trend analysis prototype
Train Delay Prediction & Monitoring	Delay predict dashboard vi.

Risk Mitigation



Risk
Limited real Sri Lankan railway data
ML prediction accuracy may be low
Real-time train tracking or delay
Ticket fraud patterns may continue

Commercialization & Sustainability

Customer Persona

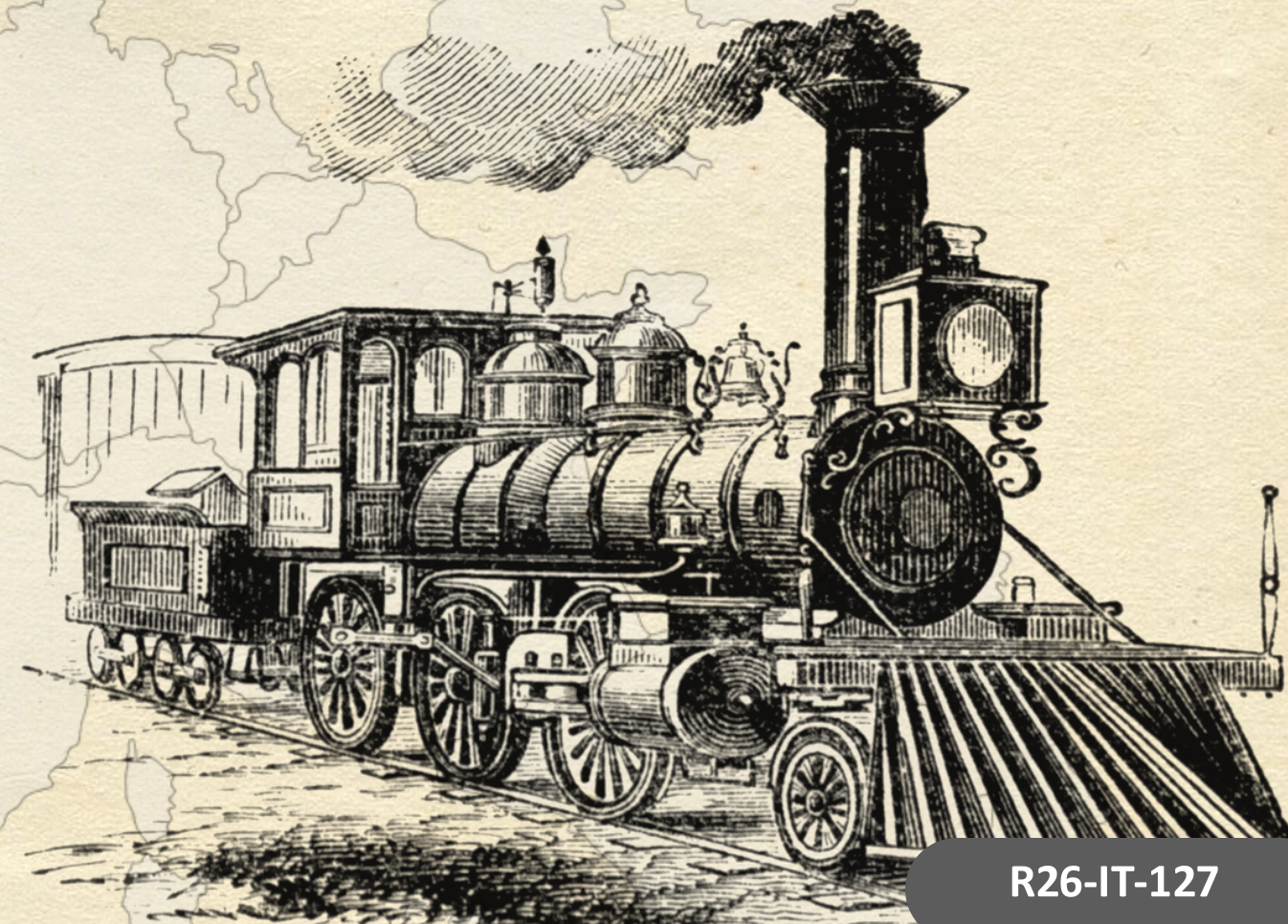
- Sri Lanka Railway Department
- Railway station managers and operators
- Train scheduling and operation teams
- Railway ticket inspectors
- Government transport authorities
- Smart transportation technology providers

Market Opportunity

- Sri Lankan railway sector needs digital transformation
- High demand for smarter passenger demand forecasting
- Need for real-time train tracking and delay prediction
- Requirement to reduce ticket fraud and revenue loss
- Opportunity to improve train scheduling and capacity management

User Benefits

- Better passenger demand prediction
- Improved train scheduling and seat allocation
- Real-time train tracking and delay updates
- Reduced ticket fraud through AI-based risk detection
- Improved operational efficiency and passenger satisfaction



Thank You

By: R26-IT-127

